

Competition Section

Physical World

GLIMPSES

- Science.** Science is the systematised and organised knowledge about the various natural phenomena which is obtained by careful experimentation, keen observation, and accurate reasoning. It has two main branches, (a) physical sciences, (b) biological sciences.
 - Physical sciences.** These sciences deal with the properties and behaviour of non-living things.
 - Biological sciences.** These sciences deal with the behaviour of living things.
- Physics.** The word physics comes from a Greek word which means nature. Physics is the study of the natural laws and their manifestation in the natural phenomena. It deals with the concepts of space, time, motion, matter, energy, radiation, etc.
- Chemistry.** Chemistry is the study of every substance, its structure, its composition and changes in which it takes part.
- Scientific attitude.** Scientific attitude requires a flexible, open-minded approach towards solving problems in which other important points of view are not neglected.
- Scientific method.** The step by step approach used by a scientist in studying natural phenomena and establishing laws which govern these phenomena, is called scientific method. It involves the steps like (i) experimentation and observation, (ii) formation of hypothesis, (iii) verification of hypothesis and (iv) theoretical predictions.
- Scientific theory.** Theory is the name given to a set of a limited number of laws in terms of which the behaviour of a physical system can be explained.
- Physics is an exact science.** It is because of high precision and accuracy obtainable in the measurement of physical quantities.
- Physics is a basic science.** Because of its important role played in the development of life science, medicine, technology and industry, physics is considered as the most basic of all sciences.
- Physics and mathematics.** Mathematics has proved to be the most important tool in the development of physics.
- Mechanics.** It deals with the slow motion or equilibrium of material bodies.
- Optics.** It deals with the nature and propagation of light.
- Thermodynamics.** It deals with the change in internal energy, temperature, entropy etc., of the macroscopic systems through external work and heat.
- Electrodynamics.** It deals with electric and magnetic phenomena associated with charged and magnetic bodies.
- Quantum mechanics.** It deals with the mechanical behaviour of atoms, molecules and nuclei.
- Relativity.** It deals with the particles having speeds comparable to the speed of light. It is the theory of invariance in nature.
- Physics, technology and society.** Sometimes, physics generates new technology. At other times technology gives rise to new physics. Both have a direct impact on society.
- Scope of physics.** The scope of physics is truly vast. It covers a wide range of magnitudes of physical quantities such as mass, length, time, energy, etc.
- Basic quests in physics.** The two basic quests in modern physics are : (i) unification and (ii) reductionism. These have resulted in great advances in physics.

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19. **Fundamental forces in nature.** There are four fundamental forces which govern both macroscopic and microscopic phenomena. These are : (i) gravitational force, (ii) electromagnetic force, (iii) strong nuclear force and (iv) weak nuclear force.

The relative strengths of these forces are

$$F_G : F_W : F_E : F_S = 1 : 10^{-25} : 10^{-36} : 10^{-38}.$$

20. **Gravitational force.** It is the force of mutual attraction between two bodies by virtue of their masses. It obeys Newton's law of gravitation. It is caused by the exchange particles called *gravitons*.
21. **Electromagnetic force.** It is the force due to interaction between two moving charges. It is caused by exchange of photons (γ) between two charged particles.
22. **Strong nuclear force.** It is the strongest attractive interaction which binds together the protons and neutrons in a nucleus. Electrons do not experience this force.
23. **Weak nuclear force.** It is the force that appears only between elementary particles involved in a

nuclear process such as the β -decay of a nucleus. It is caused by exchange of vector bosons.

24. **Isolated system.** Any system on which no external force acts is called isolated system. Universe is the most ideal isolated system possible.
25. **Law of conservation of energy.** It states that the total energy of an isolated system remains constant. It follows from the nature's symmetry called homogeneity of time.
26. **Law of conservation of linear momentum.** It states that if no external force acts on a system, then its total linear momentum remains constant. It follows from nature's symmetry called homogeneity of space.
27. **Law of conservation of angular momentum.** It states that if no external torque acts on a system, then its angular momentum remains constant. It follows from nature's symmetry called isotropy of space.
28. **Law of conservation of charge.** It states that the total charge of an isolated system remains constant.

Multiple Choice Questions

1. A supposition that is put forward as a probable solution to a natural phenomenon is called a

- (a) hypothesis (b) theory
(c) law (d) relief.

2. A thought experiment in physics is one

- (a) which is theoretically possible but experimentally not feasible.
(b) which is neither theoretically nor experimentally feasible.
(c) which is performed by non-physicist.
(d) none of the above.

3. "Nature is pleased with simplicity and effects not the pump of superfluous causes". It has been said by the great scientist

- (a) Einstein (b) Dirac
(c) Russel (d) Newton.

4. The book titled 'Principia' was written by

- (a) Maxwell (b) Einstein
(c) Newton (d) Galileo. [AFMC 01]

5. "It is more important to have beauty in the equations of physics than to have them agree with experiments".

The scientist who made this statement is

- (a) Dirac (b) Maxwell
(c) Faraday (d) Raman.

6. "The most incomprehensible thing about the world is that it is comprehensible." This statement was made by the scientist

- (a) Aristotle (b) Newton
(c) Galileo (d) Einstein.

7. The country which awards the prestigious Nobel prize is

- (a) U.S.A. (b) U.K.
(c) Sweden (d) France.

8. An Indian scientist who won Nobel prize for physics is

- (a) Sir J.C. Bose (b) H.J. Bhaba
(c) M.N. Saha (d) Sir C.V. Raman.

[AFMC 01]

9. The man who has won Nobel Prize twice in physics is

- (a) Einstein (b) Bardeen
(c) Heisenberg (d) Faraday.

[AFMC 01]

Units and Measurements

GLIMPSES

- Physical quantities.** All those quantities which can be measured directly or indirectly and in terms of which the laws of physics can be expressed are called *physical quantities*.
- Physical unit.** The standard amount of a physical quantity chosen to measure the physical quantity of the same kind is called a *physical unit*.
Measure of a physical quantity = Numerical value of the quantity $\times$ size of the unit = nu
 - Relationship between the numerical value and the size of the unit.** The numerical value (n) of a physical quantity is inversely proportional to the size of the unit.
$$n \propto \frac{1}{u} \quad \text{or} \quad nu = \text{constant} \quad \text{or} \quad n_1 u_1 = n_2 u_2$$
- Fundamental quantities.** The physical quantities which can be treated as independent of other physical quantities and are not usually defined in terms of other physical quantities are called *fundamental quantities*. The seven fundamental quantities are mass, length, time, electric current, temperature, luminous intensity and amount of substance.
- Derived quantities.** The physical quantities whose defining operations are based on other physical quantities are called derived quantities.
- Fundamental units.** The physical units which can neither be derived from one another, nor they can be further resolved into more simpler units are called *fundamental units*. The units of length, mass, time, etc. are fundamental units.
- Derived units.** All other units which can be expressed in terms of fundamental units are called *derived units*.
- System of units.** A complete set of units which is used for measuring all kinds of fundamental and derived quantities is called a *system of units*.
 - The CGS or the metric system.** In this system the fundamental units of length, mass and time are centimetre, gram and second respectively.
 - The FPS or the British system.** In this system the fundamental units of length, mass and time are foot, pound and second respectively.
 - The MKS system.** In this system the fundamental units of length, mass and time are metre, kilogram and second respectively.
 - The SI.** SI is the abbreviation for *Système Internationale d'Unités*, which is the French equivalent for *international system of units*. In this system the fundamental units of length, mass, time, electric current, temperature, luminous intensity and amount of substance are metre, kilogram, second, ampere, kelvin, candela and mole, respectively.

8. The seven basic SI units.

- Metre (m).** One metre is defined as the length of the path travelled by light in vacuum during a time interval of $1/299,792,458$ of a second (1983).
- Kilogram (kg).** It is the mass of a platinum-iridium cylinder (90% Pt and 10% Ir) whose height is equal to its diameter (each = 3.9 cm) preserved at International Bureau of Weights and Measures at Sevres (1889).
- Second (s).** One second is defined as the duration of 9,192,631,770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the cesium-133 atom (1967).
- Ampere (A).** One ampere is that constant current which, if maintained in two straight parallel conductors of infinite length, of negligible circular cross-section, and placed 1 metre

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apart in vacuum, would produce between these conductors a force equal to 2×10^{-7} newton per metre of length (1948).

- (v) **Kelvin (K).** One kelvin is the fraction $1/273.16$ of the thermodynamic temperature of the triple point of water (1967).
- (vi) **Candela (cd).** Candela is the luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency 540×10^{12} hertz and that has a radiant intensity in that direction of $1/683$ watt per steradian (1979).
- (vii) **Mole (mol).** One mole is the amount of substance of a system, which contains as many elementary entities as there are atoms in 0.012 kilogram of carbon-12 (1971).

9. The two supplementary SI units.

- (i) **Radian (rad).** It is defined as the plane angle subtended at the centre of a circle by an arc equal in length to the radius of the circle.

$$\theta \text{ (in radians)} = \frac{\text{Arc}}{\text{Radius}} = \frac{l}{r}$$

- (ii) **Steradian (sr).** It is the solid angle subtended at the centre of a sphere by a surface of the sphere equal in area to that of a square, having each side equal to the radius of the sphere.

$$\Omega \text{ (in steradian)} = \frac{\text{Surface area}}{\text{Radius}^2}$$

- 10. **Coherent system.** A coherent system of units is a system based on a certain set of fundamental units from which all derived units are obtained by multiplication or division without introducing any numerical factor.

- 11. **Advantages of SI.** It is a metric, rational, coherent and internationally acceptable system of units which covers all branches of physics.

12. Some Practical Units.

- (i) 1 fermi = 10^{-15} m
- (ii) 1 angstrom ($\text{\AA}$) = 10^{-10} m
- (iii) 1 nanometre (nm) = 10^{-9} m
- (iv) 1 micron (μm) = 10^{-6} m
- (v) 1 light year (ly) = 9.46×10^{15} m
- (vi) 1 astronomical unit (AU) = 1.496×10^{11} m
- (vii) 1 parallaxic second (parsec) = 3.08×10^{16} m
- (viii) 1 barn = 10^{-28} m²
- (ix) 1 tonne = 1000 kg
- (x) 1 quintal = 100 kg
- (xi) 1 slug = 14.57 kg
- (xii) 1 atomic mass unit (amu) = 1.66×10^{-27} kg

- (xiii) 1 solar year = 365.25 average solar days
= 366.25 sidereal days

- (xiv) 1 lunar month = 27.3 days

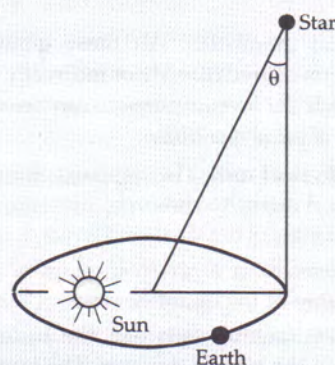
- (xv) 1 shake = 10^{-8} s

- (xvi) 1 Torr = 1 mm of Hg

- (xvii) 1 bar = 1 atmospheric pressure
= 760 mm of Hg = 10^5 Nm^{-2} .

- 13. **Order of magnitude.** The order of magnitude of a physical quantity is that power of ten which is closest to its magnitude. To determine the order of magnitude of a number N , we express it as $N = n \times 10^x$. If $0.5 < n \leq 5$, then x will be the order of magnitude of N .

- 14. **Annual parallax.** It is the angle (θ) at which the semi-major axis of the earth's orbit perpendicular to the star's direction is seen from the star.



- 15. **Indirect methods for measuring long distances.**

- (i) **Reflection or echo method.** The distance (S) of a hill can be determined by noting the time interval t of sending a sound wave towards the hill and receiving back its echo.

$$S = \frac{c \times t}{2},$$

where c = Speed of sound waves

The same principle is used in radars and sonars.

- (ii) **Triangulation method.** The height (h) of an accessible object like a tower or a tree can be found by measuring the angle θ subtended by the object at the observation point. If x is the distance of the observation point from the foot of the object, then

$$h = x \tan \theta$$

To find the height of an inaccessible object like a hill, its angles of elevation θ_1 and θ_2 are measured at two points separated by distance d .

$$h = \frac{d}{\cot \theta_2 - \cot \theta_1}.$$

- (iii) **Parallax method.** By measuring the angle θ (called parallax angle or parallaxic angle) subtended

by the astronomical object at two locations on the earth separated by a large distance b (called basis), the distance S of the object can be determined.

$$\theta = \frac{\text{Arc}}{\text{Radius}} = \frac{b}{S} \quad \text{or} \quad S = \frac{\text{Basis}}{\text{Parallactic angle}} = \frac{b}{\theta}$$

(iv) **Size of an astronomical object.** The angle θ (called angular diameter) subtended by the diameter of the planet is measured at a point on the surface of the earth. Knowing the distance S of the planet from the earth, its diameter D can be determined.

$$\theta = \frac{\text{Arc}}{\text{Radius}} = \frac{D}{S} \quad \text{or} \quad D = S \times \theta$$

Linear diameter = Distance $\times$ Angular diameter.

(v) **Copernicus method.** This method is used to determine the distance of an inferior planet from the sun. The angle formed at the earth between the earth-planet direction and the earth-sun direction is called the planet's elongation (ε).

Distance of inferior planet from the sun,

$$r_{ps} = \sin \varepsilon \cdot r_{es} = \sin \varepsilon \text{ AU}$$

Distance of inferior planet from the earth,

$$r_{pe} = \cos \varepsilon \cdot r_{es} = \cos \varepsilon \text{ AU}.$$

(vi) **By applying Kepler's third law.** If T_1 and T_2 be the periods of revolution of two planets and a_1 and a_2 be their semi-major axes, then

$$\frac{a_2^3}{a_1^3} = \frac{T_2^2}{T_1^2}$$

$$\text{or} \quad a_2 = \left[\frac{T_2}{T_1} \right]^{2/3} \cdot a_1$$

16. **Molar Volume.** One mole of every gas occupies a volume of 22.4 litres at S.T.P. This is called molar volume of the gas. It contains an Avogadro's number ($N = 6.023 \times 10^{23}$) of molecules.

17. **Indirect methods for measuring small distances.**

(i) **By using Avogadro's hypothesis.** According to Avogadro's hypothesis, the actual volume occupied by the atoms in one gram of a substance is two-third of the volume of one gram of the substance.

Radius of atom,

$$r = \left[\frac{M}{2\pi N \rho} \right]^{1/3}$$

where N , M and ρ are the Avogadro's number, molecular weight and density of the substance.

(ii) **Molecular size of oleic acid.** A large and very thin film of oil is formed on water surface and its area is measured.

Thickness of oil film,

$$t = \frac{\text{Volume of the film}}{\text{Area of the film}}$$

Assuming that the film is of one molecular thickness, then t is approximately the size of oil molecule.

19. **Mass.** The mass of a body is the quantity of matter contained in it. Its SI unit is kg.

20. **Weight.** The weight of a body is the force with which a body is attracted towards the centre of the earth. $W = mg$.

21. **Inertial mass.** The mass of a body which determines its inertia in translatory motion is called its inertial mass. It is measured from Newton's second law of motion.

$$m_i = \frac{F}{a}$$

22. **Gravitational mass.** The mass of a body which determines the gravitational pull upon it due to the earth is called its gravitational mass. It is measured by using Newton's law of gravitation.

$$m_g = \frac{FR^2}{GM}$$

23. **Dimensions of the derived quantity.** These are the powers to which the fundamental units of mass, length and time must be raised in order to represent a derived quantity completely.

24. **Dimensional formula.** It is an expression which shows how and which of the fundamental units of mass, length and time occur in the derived unit of a physical quantity.

25. **Dimensional equation.** The equation which expresses a physical quantity in terms of the fundamental units of mass, length and time, is called dimensional equation.

26. **Principle of homogeneity of dimensions.** This principle states that a physical equation will be dimensionally correct if the dimensions of all the terms occurring on both sides of the equation are the same.

27. **Uses of dimensional equations:**

(i) To convert a physical quantity from one system of units to another.

Suppose a physical quantity has dimensional formula $M^a L^b T^c$. Let n_1 and n_2 be its numerical values when the units are u_1 and u_2 . Then

$$n_1 u_1 = n_2 u_2$$

$$n_1 [M_1^a L_1^b T_1^c] = n_2 [M_2^a L_2^b T_2^c]$$

$$\therefore n_2 = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$$

- (ii) To check the correctness of a given physical relation.
- (iii) To derive a relationship between different physical quantities.
28. **Significant figures.** The significant figures are normally those digits in a measured quantity which are known reliably plus one additional digit that is uncertain.
29. **Rules for counting the number of significant figures in a measured quantity :**
- All non-zero digits are significant.
 - All zeros between two non-zero digits are significant.
 - All zeros to the right of a non-zero digit but to left of an understood decimal point are not significant. But such zeros are significant if they come from a measurement.
 - All zeros to the right of a non-zero digit but to the left of a decimal point are significant. For example, 648700. has six significant figures.
 - All zeros to the right of a decimal point are significant.
 - All zeros to the right of a decimal point but to the left of a non-zero digit are not significant. Single zero conventionally placed to the left of the decimal point is not significant.
 - The number of significant figures does not depend on the system of units.
30. **Significant figures in the sum or difference of two numbers.** In addition or subtraction the result should be reported to the same number of decimal places as that of the number with minimum number of decimal places.
31. **Significant figures in the product or quotient of two numbers.** In multiplication or division, the result should be reported to the same number of significant figures as that of the number with minimum of significant figures.
32. **Rules for rounding off a measurement :**
- If the digit to be dropped is less than 5, then the preceding digit is left unchanged.
 - If the digit to be dropped is greater than 5, then the preceding digit is increased by 1.
 - If the digit to be dropped is 5 followed by non-zero digits, then the preceding digit is increased by 1.
 - If the digit to be dropped is 5, then the preceding digit is left unchanged if it is even.
 - If the digit to be dropped is 5, then the preceding digit is increased by 1 if it is odd.
33. **Accuracy and precision of a measurement.** Accuracy refers to the closeness of a measurement to the true value of the physical quantity. Precision refers to the resolution or the limit to which the quantity is measured.
34. **Error in a measurement.** It is the difference between the measured value and the true value of a physical quantity. It gives an indication of the limits within which the true value may lie.
35. **Elimination of random error.** The errors which occur irregularly and at random, in magnitude and direction, are called random errors. To eliminate random error, a large number of readings are taken and their arithmetic mean is taken as the true value.
- $$\bar{a} = \frac{a_1 + a_2 + a_3 + \dots + a_n}{n} = \frac{1}{n} \sum_{i=1}^n a_i$$
36. **Absolute error.** The magnitude of the difference between the true value and the measured value is called absolute error.
- Such errors are given by
- $$\Delta a_1 = \bar{a} - a_1 \quad \Delta a_2 = \bar{a} - a_2$$
- $$\Delta a_3 = \bar{a} - a_3 \quad \Delta a_n = \bar{a} - a_n$$
37. **Mean absolute error.** The arithmetic mean of the positive magnitudes of all the absolute errors is called mean absolute error
- $$\Delta \bar{a} = \frac{|\Delta a_1| + |\Delta a_2| + \dots + |\Delta a_n|}{n} = \frac{1}{n} \sum_{i=1}^n |\Delta a_i|$$
38. **Relative error.** It is the ratio of the mean absolute error to the true value.
- $$\delta a = \frac{\Delta \bar{a}}{\bar{a}}$$
39. **Percentage error.** The relative error expressed in percent is called percentage error.
- $$\text{Percentage error} = \frac{\Delta \bar{a}}{\bar{a}} \times 100\%$$
40. **Error combination in a sum or a difference.** When two quantities are added or subtracted, the absolute error in the final result is the sum of the absolute errors associated with the individual quantities.
- $$\Delta Z = \Delta A + \Delta B$$
41. **Error combination in a product or a quotient.** When two quantities are multiplied or divided, the fractional error in the final result is the sum of the fractional errors of the two quantities.
- $$\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

42. **Error due to the power of a measured quantity.** The fractional error in the n th power of a quantity is equal to n times the fractional error in the quantity itself.

$$\text{If } Z = A^n, \text{ then } \frac{\Delta Z}{Z} = n \frac{\Delta A}{A}$$

43. **General rule.** If $Z = \frac{A^p B^q}{C^r}$, then the maximum fractional or relative error in Z will be

$$\frac{\Delta Z}{Z} = p \frac{\Delta A}{A} + q \frac{\Delta B}{B} + r \frac{\Delta C}{C}$$

$$\% \text{ Error in } Z = \frac{\Delta Z}{Z} \times 100$$

$$= p \frac{\Delta A}{A} \times 100 + q \frac{\Delta B}{B} \times 100 + r \frac{\Delta C}{C} \times 100$$

44. **Propagation or combination of errors.** It is summarised in the adjacent table.

Operation	Formula Z	Absolute error ΔZ	Relative error $\frac{\Delta Z}{Z}$	Percentage error $100 \Delta Z / Z$
Sum	$A + B$	$\Delta A + \Delta B$	$\frac{\Delta A + \Delta B}{A + B}$	$\frac{\Delta A + \Delta B}{A + B} \times 100$
Difference	$A - B$	$\Delta A + \Delta B$	$\frac{\Delta A + \Delta B}{A - B}$	$\frac{\Delta A + \Delta B}{A - B} \times 100$
Multi- plication	$A \times B$	$A \Delta B + B \Delta A$	$\frac{\Delta A}{A} + \frac{\Delta B}{B}$	$\left(\frac{\Delta A}{A} + \frac{\Delta B}{B} \right) \times 100$
Division	$\frac{A}{B}$	$\frac{B \Delta A + A \Delta B}{B^2}$	$\frac{\Delta A}{A} + \frac{\Delta B}{B}$	$\left(\frac{\Delta A}{A} + \frac{\Delta B}{B} \right) \times 100$
Power	A^n	$n A^{n-1} \Delta A$	$n \frac{\Delta A}{A}$	$n \frac{\Delta A}{A} \times 100$
Root	$A^{1/n}$	$\frac{1}{n} A^{1/n-1} \Delta A$	$\frac{1}{n} \frac{\Delta A}{A}$	$\frac{1}{n} \frac{\Delta A}{A} \times 100$

IIT Entrance Exam

MULTIPLE CHOICE QUESTIONS WITH ONE CORRECT ANSWER

1. The equation of state for a real gas is given by

$$\left(P + \frac{a}{V^2} \right) (V - b) = RT$$

The dimensions of the constant a are

- (a) $[ML^5T^{-2}]$ (b) $[M^{-1}L^5T^{-2}]$
(c) $[ML^{-5}T^{-1}]$ (d) $[ML^5T^{-1}]$

[IIT 97; CBSE PMT 96]

2. Pressure depends on distance as $P = \frac{\alpha}{\beta} \exp\left(\frac{-\alpha z}{k\theta}\right)$,

where α, β are constants, z is distance, k is Boltzmann's constant and θ is temperature. The dimensions of β are

- (a) $[M^0L^0T^0]$ (b) $[M^{-1}L^{-1}T^{-1}]$
(c) $[M^0L^2T^0]$ (d) $[M^{-1}L^{-1}T^{-2}]$ [IIT 04]

3. The dimensions of $\left(\frac{1}{2}\right)\epsilon_0 E^2$ are

- (ϵ_0 : permittivity of free space, E : electric field)
(a) $[MLT^{-1}]$ (b) $[ML^2T^{-2}]$
(c) $[ML^{-1}T^{-2}]$ (d) $[ML^2T^{-1}]$ [IIT 2K]

4. In the formula : $X = 3YZ^2$, X and Z have dimensions of capacitance and magnetic induction respectively. What are the dimensions of Y in MKSQ system ?

- (a) $[M^{-3}L^{-1}T^3Q^4]$ (b) $[M^{-3}L^{-2}T^4Q^4]$
(c) $[M^{-2}L^{-2}T^4Q^4]$ (d) $[M^{-3}L^{-2}T^4Q^1]$ [IIT 95]

5. A quantity X is given by $\epsilon_0 L \frac{\Delta V}{\Delta t}$ where ϵ_0 is the

permittivity of the free space, L is a length, ΔV is a potential difference and Δt is a time interval. The dimensional formula for X is the same as that of

- (a) resistance (b) charge
(c) voltage (d) current [IIT 01]

6. Which of the following sets have different dimensions ?

- (a) Pressure, Young's modulus, stress
(b) Emf, potential difference, electric potential
(c) Heat, work done, energy
(d) Dipole moment, electric flux, electric field [IIT 05]

7. A cube has a side of length 1.2×10^{-2} m. Calculate its volume.

- (a) $1.7 \times 10^{-6} \text{ m}^3$ (b) $1.73 \times 10^{-6} \text{ m}^3$
(c) $1.0 \times 10^{-6} \text{ m}^3$ (d) $1.732 \times 10^{-6} \text{ m}^3$ [IIT 03]

8. A student performs an experiment for determination of $g = \left(\frac{4\pi^2 l}{T^2} \right)$. The error in length l is Δl and in time T is ΔT and n is number of times the reading is taken. The measurement of g is most accurate for

- | | Δl | ΔT | n |
|-----|------------|------------|-----|
| (a) | 5 mm | 0.2 sec | 10 |
| (b) | 5 mm | 0.2 sec | 20 |
| (c) | 5 mm | 0.1 sec | 10 |
| (d) | 1 mm | 0.1 sec | 50 |
- [IIT 06]

Competition Section

Motion in a Straight Line

GLIMPSES

1. **Mechanics.** It is the branch of physics that deals with the conditions of rest or motion of the material objects around us.
2. **Statics.** It is the branch of mechanics that deals with the study of objects at rest or in equilibrium.
3. **Kinematics.** It is the branch of mechanics that deals with the study of motion of objects without considering the cause of motion.
4. **Dynamics.** It is the branch of mechanics that deals with the study of motion of objects taking into consideration the cause of their motion.
5. **Rest.** An object is at rest if it does not change its position w.r.t. its surroundings with the passage of time.
6. **Motion.** An object is in motion if it changes its position w.r.t. its surroundings with the passage of time.
7. **Rest and motion are relative terms.** No body can exist in a state of absolute rest or of absolute motion.
8. **Point object.** If the position of an object changes by distances much greater than its own size in a reasonable duration of time, then the object may be regarded as a point object.
9. **One dimensional motion.** The motion of an object is said to be one dimensional motion if only one out of the three coordinates specifying the position of the object changes with time. In such a motion, an object moves along a straight line path.
10. **Two dimensional motion.** The motion of an object is said to be two dimensional motion if two out of the three coordinates specifying the position of the object change with time. In such a motion, the object moves in a plane.
11. **Three dimensional motion.** The motion of an object is said to be three dimensional motion if all

the three coordinates specifying the position of the object change with time. In such a motion, the object moves in space.

12. **Distance or path length.** It is the length of the actual path traversed by a body between its initial and final positions. It is a scalar quantity. Its SI unit is metre. It is always positive or zero.
13. **Displacement.** It is defined as the change in the position of an object in a fixed direction. It is given by the vector drawn from the initial position to the final position of the object. It is a vector quantity. It can be positive, negative or zero. Its SI unit is metre.
The magnitude of displacement is less than or equal to the actual distance travelled by the object in the given time interval.
14. **Speed.** It is rate of change of position of a body in any direction.

$$\text{Speed} = \frac{\text{Distance travelled}}{\text{Time taken}}$$

Speed is a scalar quantity. Its SI unit is ms^{-1} and its dimensional formula is $[\text{M}^0 \text{L}^1 \text{T}^{-1}]$.

15. **Uniform speed.** An object is said to be moving with uniform speed if it covers equal distances in equal intervals of time, however small these time intervals may be.
16. **Variable speed.** An object is said to be moving with variable speed if it covers unequal distances in equal intervals of time.
17. **Average speed.** It is equal to the total distance travelled by the object divided by the total time taken to cover that distance.

$$\text{Average speed} = \frac{\text{Total distance travelled}}{\text{Total time taken}}$$

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18. **Instantaneous speed.** The speed of an object at any particular instant of time or at a particular point of its path is called the instantaneous speed of the object.

$$\text{Instantaneous speed, } v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

19. **Velocity.** It is the rate of change of position of an object in a particular direction. It is equal to the displacement covered by a body per unit time.

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}}$$

Velocity is a vector quantity. Its SI unit is ms^{-1} and dimensional formula is $[\text{M}^0 \text{L}^1 \text{T}^{-1}]$.

20. **Uniform velocity.** A body is said to be moving with uniform velocity if it covers equal displacements in equal intervals of time, however small these time intervals may be.
21. **Variable velocity.** A body is said to be moving with variable velocity if either its speed changes or direction of motion changes or both change with time.
22. **Average velocity.** It is equal to the net displacement covered divided by the total time taken.

$$v_{av} = \frac{\text{Net displacement}}{\text{Total time taken}} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta x}{\Delta t}$$

23. **Instantaneous velocity.** The velocity of an object at a particular instant of time or at a particular point of its path is called its instantaneous velocity.

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{x}}{\Delta t} = \frac{d\vec{x}}{dt}$$

24. **Uniform motion.** An object is said to be in uniform motion if it covers equal distances in equal intervals of time, however small these intervals may be, in the same fixed direction.
25. **Displacement in uniform motion.** If at time $t = 0$, the displacement is x_0 , then
- $$x = x_0 + vt \quad \text{or} \quad x - x_0 = vt \quad \text{or} \quad s = vt.$$
26. **Non-uniform motion.** A body is said to be in non-uniform motion if its velocity changes with time.
27. **Acceleration.** The rate of change of velocity of an object is called its acceleration.

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{Time taken}}$$

Acceleration is a vector quantity. Its SI unit is ms^{-2} and its dimensional formula is $[\text{M}^0 \text{L}^1 \text{T}^{-2}]$.

28. **Uniform acceleration.** If the velocity of an object changes by equal amounts in equal intervals of time, however small these intervals may be, then the object is said to move with uniform or constant acceleration.

29. **Variable acceleration.** If the velocity of an object changes by unequal amounts in equal intervals of time, then the object is said to be in variable acceleration.

30. **Average acceleration.** The average acceleration of an object between two points on the path of its motion is defined as ratio of the change in velocity to the total time interval in which that change has taken place.

$$a_{av} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta v}{\Delta t}$$

31. **Instantaneous acceleration.** The acceleration of an object at a given instant of time or at a given point of its motion is called its instantaneous acceleration.

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d^2 \vec{x}}{dt^2}$$

32. **Positive acceleration.** If the velocity of an object increases with time, its acceleration is positive.
33. **Negative acceleration.** If the velocity of an object decreases with time, its acceleration is negative. Negative acceleration is also called *retardation* or *deceleration*.

34. Equations of motion for constant acceleration

- (a) *In conventional form.* Let u be the initial velocity of a particle, a the uniform acceleration, v its velocity after time t and s is the distance travelled in time t , then the following equations hold good :

$$(i) v = u + at \quad (ii) s = \frac{u + v}{2} \times t$$

$$(iii) s = ut + \frac{1}{2} at^2 \quad (iv) v^2 - u^2 = 2as$$

$$(v) s = vt - \frac{1}{2} at^2 \quad (vi) s_{nth} = u + \frac{a}{2} (2n - 1)$$

where s_{nth} = the distance travelled in n^{th} second.

- (b) *In cartesian form.* Suppose a particle moves with uniform acceleration a along X-axis. Let x_0 and x be its position co-ordinates and v_0 and v be its velocities at times $t = 0$ and t respectively. Then the following equations hold good :

$$(i) v = v_0 + at$$

$$(ii) x = x_0 + v_0 t + \frac{1}{2} at^2 \quad \text{or} \quad s = v_0 t + \frac{1}{2} at^2$$

$$(iii) v^2 = v_0^2 + 2a(x - x_0) = v_0^2 + 2as$$

$$(iv) s_{nth} = v_0 + \frac{a}{2} (2n - 1)$$

35. **Free fall.** In the absence of air resistance, all bodies fall with the same acceleration near the surface of the earth. This motion of a body falling towards the earth from a small height is called free fall. The acceleration with which a body falls is called acceleration due to gravity and is denoted by g . Near the surface of the earth, $g = 9.8 \text{ ms}^{-2}$.

- 36. Motion under gravity.** For a freely falling body, the following equations of motion hold good :

$$(i) v = u + gt \quad (ii) s = ut + \frac{1}{2}gt^2$$

$$(iii) v^2 - u^2 = 2gs$$

When a body falls freely under the action of gravity, its velocity increases and the value of g is taken *positive*.

When a body is thrown vertically upward, its velocity decreases and the value of g is taken *negative*.

For a body thrown vertically upward with initial velocity u , we have

- (i) Maximum height reached, $h = \frac{u^2}{2g}$
- (ii) Time of ascent = Time of descent = $\frac{u}{g}$
- (iii) Total time of flight to come back to the point of projection = $\frac{2u}{g}$
- (iv) Velocity of fall at the point of projection = u
- (v) Velocity attained by a body dropped through height h , $v = \sqrt{2gh}$.

- 37. Position-time graph.** It is the graph between the time t and position x of a particle relative to a fixed origin. Its slope at any point gives the instantaneous velocity at that point.

- (i) For a stationary object, the position-time graph is a straight line parallel to the time-axis.
- (ii) For a body in uniform motion, the position-time graph is a straight line inclined to the time-axis.
- (iii) For uniformly accelerated motion, the position-time graph is a parabola.

- 38. Velocity-time graph.** It is a graph of time versus velocity. Its slope at any point gives the acceleration at the corresponding instant. Distance

covered in time t equals area under the velocity-time graph bounded by the time-axis.

- (i) For uniform motion, the velocity-time graph is a straight line parallel to the time-axis.
- (ii) For uniform acceleration, the velocity-time graph is a straight line inclined to the time-axis.
- (iii) For variable acceleration, the velocity-time graph is a curve.

- 39. Relative velocity.** The relative velocity of an object B with respect to object A when both are in motion is the rate of change of position of object B with respect to object A .

Relative velocity of object B w.r.t. object A ,

$$\vec{v}_{BA} = \vec{v}_B - \vec{v}_A$$

Relative velocity of object A w.r.t. object B ,

$$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$$

When both the objects A and B move in the same direction,

$$v_{AB} = v_A - v_B$$

When the object B moves in the opposite direction of A ,

$$v_{AB} = v_A + v_B$$

When v_A and v_B are inclined to each other at angle θ ,

$$v_{AB} = \sqrt{v_A^2 + v_B^2 + 2v_A v_B \cos(180^\circ - \theta)}$$

$$= \sqrt{v_A^2 + v_B^2 - 2v_A v_B \cos \theta}$$

If v_{AB} makes angle β with v_A , then

$$\tan \beta = \frac{v_B \sin \theta}{v_A - v_B \cos \theta}$$

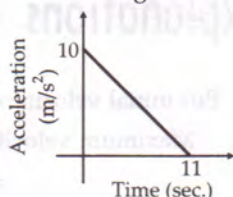
IIT Entrance Exam

MULTIPLE CHOICE QUESTIONS WITH ONE CORRECT ANSWER

- 1.** A body starts from rest at time $t=0$, the acceleration-time graph is shown in figure. The maximum velocity attained by the body will be

- (a) 110 m/s (b) 55 m/s
(c) 650 m/s (d) 550 m/s

[IIT 04]



- 2.** A small block slides, without friction, down an inclined plane starting from rest. Let s_n be the distance travelled from $t=(n-1)$ to $t=(n)$.

Then $\frac{s_n}{s_{n+1}}$ is

- (a) $\frac{2n-1}{2n}$ (b) $\frac{2n+1}{2n-1}$
(c) $\frac{2n-1}{2n+1}$ (d) $\frac{2n}{2n+1}$

[IIT 04]

Competition Section

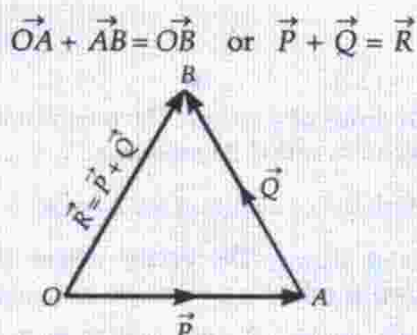
Motion in a Plane

GLIMPSES

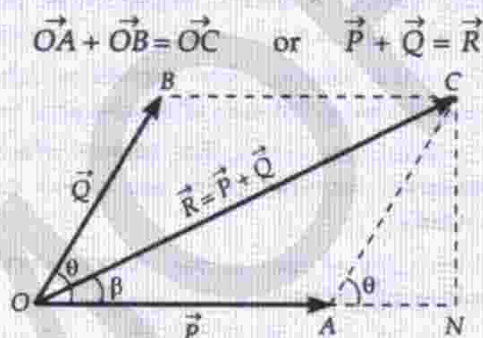
- Scalars.** The physical quantities which have only magnitude and no direction are called scalars e.g., mass, length, time, speed, work, power, etc.
- Vectors.** The physical quantities which have both magnitude and direction are called vectors e.g., displacement, velocity, acceleration, force, momentum, etc.
- Representation of a vector.** A vector is represented by a straight line with an arrowhead over it. The length of the line gives the magnitude and the arrowhead gives the direction of the vector.
- Position vector.** A vector which gives position of an object with reference to the origin of a coordinate system is called position vector.
- Displacement vector.** It is that vector which tells how much and in which direction an object has changed its position in a given time interval.
- Polar vectors.** These are the vectors which have a starting point or a point of application e.g., displacement, force, velocity, etc.
- Axial vectors.** The vectors which represent rotational effect and act along the axis of rotation in accordance with right hand screw rule are called axial vectors e.g., torque, angular momentum, etc.
- Equal vectors.** Two vectors are said to be equal if they have the same magnitude and direction.
- Negative vector.** The negative of a vector is defined as another vector having the same magnitude but having an opposite direction.
- Zero vector.** A vector having zero magnitude and an arbitrary direction is called a zero or null vector.
- Collinear vectors.** The vectors which either act along the same line or along parallel lines are called collinear vectors.
- Coplanar vectors.** The vectors which act in the same plane are called coplanar vectors.
- Modulus of a vector.** The magnitude or length of a vector is called its modulus.
Modulus of vector $\vec{A} = |\vec{A}| = A$
- Fixed vector.** The vector whose initial vector is fixed is called a *fixed vector* or *localised vector*.
- Unit vector.** A unit vector is a vector of unit magnitude drawn in the direction of a given vector. A unit vector in the direction of $\vec{A}$ is given by
$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}$$
- Free vector.** A vector whose initial point is not fixed is called a *free vector* or *non-localised vector*.
- Co-initial vectors.** The vectors which have the same initial point are called co-initial vectors.
- Co-terminus vectors.** The vectors which have the common terminal point are called co-terminus vectors.
- Properties of zero vector.** A zero vector has the following properties:
$$\vec{A} + \vec{O} = \vec{A}; \quad \lambda \vec{O} = \vec{O}; \quad 0 \vec{A} = \vec{O}$$
- Multiplication of vector by a real number.** When a vector $\vec{A}$ is multiplied by a real number λ , we get another vector $\lambda \vec{A}$. The magnitude of $\lambda \vec{A}$ is λ times the magnitude of $\vec{A}$. If λ is positive, then the direction of $\lambda \vec{A}$ is same as that of $\vec{A}$. If λ is negative, then the direction of $\lambda \vec{A}$ is opposite to that of $\vec{A}$.
- Multiplication of a vector by a scalar.** When a vector $\vec{A}$ is multiplied by a scalar λ , which has

certain units, the units of $\lambda \vec{A}$ are obtained by multiplying the units of $\vec{A}$ by the units of λ .

22. **Composition of vectors.** The resultant of two or more vectors is that single vector which produces the same effect as the individual vectors together would produce. The process of adding two or more vectors is called the composition of vectors.
23. **Triangle law of vector addition.** If two vectors can be represented both in magnitude and direction by the two sides of a triangle taken in the same order, then their resultant is represented completely both in magnitude and direction by the third side of the triangle taken in the opposite order. In the figure below,



24. **Parallelogram law of vector addition.** If two vectors acting simultaneously at a point can be represented both in magnitude and direction by the two adjacent sides of a parallelogram, then their resultant is represented completely both in magnitude and direction by the diagonal of the parallelogram passing through that point. In the figure,



The magnitude of the resultant $\vec{R}$ is given by

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$$

where θ is the angle between $\vec{P}$ and $\vec{Q}$. If $\vec{R}$ makes angle β with $\vec{P}$, then

$$\tan \beta = \frac{Q \sin \theta}{P + Q \cos \theta}$$

25. **Polygon law of vector addition.** If a number of vectors are represented both in magnitude and direction by the sides of an open polygon taken in the same order, then their resultant is represented

both in magnitude and direction by the closing side of the polygon taken in opposite order.

26. **Properties of vector addition :**

- (i) Vectors representing physical quantities of same nature can only be added.
(ii) Vector addition is commutative.

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}$$

- (iii) Vector addition is associative.

$$(\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C})$$

27. **Subtraction of vectors.** The subtraction of a vector $\vec{B}$ from vector $\vec{A}$ is defined as the addition of vector $-\vec{B}$ to $\vec{A}$. Thus $\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$.

28. **Resolution of a vector.** The process of splitting a vector into two or more vectors is known as resolution of the vector. The vectors into which the given vector is splitted are called component vectors. A vector $\vec{A}$ can be resolved into components along two given vectors $\vec{a}$ and $\vec{b}$ lying in the same plane in one and only one way :

$$\vec{A} = \lambda \vec{a} + \mu \vec{b}$$

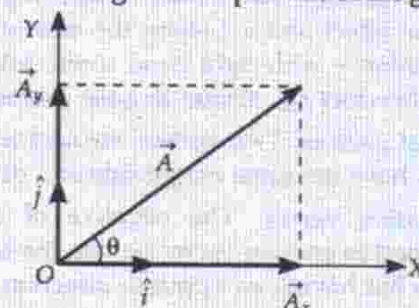
where λ and μ are real numbers.

29. **Orthogonal triad of unit vectors : Base vectors.**

The unit vectors $\hat{i}, \hat{j}, \hat{k}$ are vectors of unit magnitude and point in the direction of the x -, y -, and z -axes, respectively in a right-handed coordinate system. These are collectively known as the orthogonal triad of unit vectors or base vectors.

$$|\hat{i}| = |\hat{j}| = |\hat{k}| = 1$$

30. **Rectangular components of a vector.** When a vector is resolved along two mutually perpendicular directions, components so obtained are called rectangular components of the given vector.



As shown in figure, if $\vec{A}$ makes angle θ with X -axis and $\vec{A}_x$ and $\vec{A}_y$ are the rectangular components of $\vec{A}$ along X - and Y -axis respectively, then

$$\vec{A} = \vec{A}_x + \vec{A}_y = A_x \hat{i} + A_y \hat{j}$$

Also, $A_x = A \cos \theta$, $A_y = A \sin \theta$
 $A = \sqrt{A_x^2 + A_y^2}$ and $\tan \theta = \frac{A_y}{A_x}$.

Any vector in three dimensions can be expressed in terms of its rectangular components as

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

Its magnitude, $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$.

- 31. Scalar or dot product.** The scalar or dot product of two vectors $\vec{A}$ and $\vec{B}$ is defined as the product of the magnitudes of $\vec{A}$ and $\vec{B}$ and cosine of the angle θ between them. Thus

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta = AB \cos \theta$$

It can be positive, negative or zero depending upon the value of θ .

- 32. Geometrical interpretation of scalar product.** The scalar product of two vectors can be interpreted as the product of magnitude of one vector and component of the other vector along the first vector.

- 33. Properties of dot product of two vectors :**

- (i) For parallel vectors,
 $\theta = 0^\circ$, $\cos \theta = 1$, $\vec{A} \cdot \vec{B} = AB$
- (ii) For antiparallel vectors,
 $\theta = 180^\circ$, $\cos \theta = -1$, $\vec{A} \cdot \vec{B} = -AB$
- (iii) For perpendicular vectors,
 $\theta = 90^\circ$, $\cos \theta = 0$, $\vec{A} \cdot \vec{B} = 0$
- (iv) $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$ (Commutative law)
- (v) $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$ (Distributive law)
- (vi) $\vec{A} \cdot \vec{A} = A^2$
- (vii) $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$
- (viii) $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$

- 34. Dot product in Cartesian co-ordinates.** For

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k} \text{ and } \vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k},$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Angle θ between $\vec{A}$ and $\vec{B}$ is given by

$$\begin{aligned} \cos \theta &= \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|} \\ &= \frac{A_x B_x + A_y B_y + A_z B_z}{\sqrt{A_x^2 + A_y^2 + A_z^2} \sqrt{B_x^2 + B_y^2 + B_z^2}} \end{aligned}$$

- 35. Examples of dot product.**

- (i) Work done, $W = \vec{F} \cdot \vec{S}$
- (ii) Instantaneous power, $P = \vec{F} \cdot \vec{v}$

- 36. Vector or cross product.** For two vectors $\vec{A}$ and $\vec{B}$ inclined at an angle θ , the vector or cross product is defined as

$$\vec{A} \times \vec{B} = AB \sin \theta \hat{n}$$

where $\hat{n}$ is a unit vector perpendicular to the plane of $\vec{A}$ and $\vec{B}$ and its direction is that in which a right handed screw advances when rotated from $\vec{A}$ to $\vec{B}$.

- 37. Geometrical interpretation of vector product.** The magnitude of the vector product of two vectors is equal to (i) the area of the parallelogram formed by the two vectors as its adjacent sides and (ii) twice the area of the triangle formed by the two vectors as its adjacent sides.

- 38. Properties of cross product of two vectors :**

- (i) For parallel or antiparallel vectors,
 $\theta = 0^\circ \text{ or } 180^\circ$
 $\vec{A} \times \vec{B} = \vec{0}$
- (ii) $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$ (Anti-commutative law)
- (iii) $\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$ (Distributive law)
- (iv) $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{0}$
- (v) $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$
- (vi) Unit vector perpendicular to the plane of $\vec{A}$ and $\vec{B}$ is given by

$$\hat{n} = \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|}$$

- (vii) Angle θ between $\vec{A}$ and $\vec{B}$ is given by

$$\sin \theta = \frac{|\vec{A} \times \vec{B}|}{|\vec{A}| |\vec{B}|}$$

- 39. Cross product in Cartesian co-ordinates.**

$$\vec{A} \times \vec{B} = (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})$$

$$\begin{aligned} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \\ &= (A_y B_z - A_z B_y) \hat{i} - (A_x B_z - A_z B_x) \hat{j} \\ &\quad + (A_x B_y - A_y B_x) \hat{k}. \end{aligned}$$

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40. **Examples of cross product.** (i) Moment of force or torque,

$$\vec{\tau} = \vec{r} \times \vec{F}$$

- (ii) Angular momentum,

$$\vec{L} = \vec{r} \times \vec{p}$$

41. **Position and displacement vectors.** The position vector of an object in x - y plane is given by

$$\vec{r} = x\hat{i} + y\hat{j}$$

and the displacement from position $\vec{r}$ to position $\vec{r}'$ is given by

$$\Delta\vec{r} = \vec{r}' - \vec{r} = (x' - x)\hat{i} + (y' - y)\hat{j} = \Delta x\hat{i} + \Delta y\hat{j}$$

42. **Velocity vector.** If an object undergoes a displacement $\Delta\vec{r}$ in time Δt , its average velocity is given by

$$\vec{v} = \frac{\Delta\vec{r}}{\Delta t}$$

The velocity of an object at time t is the limiting value of the average velocity as Δt tends to zero. Thus

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

In component form, we have

$$\vec{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$$

where $v_x = \frac{dx}{dt}$, $v_y = \frac{dy}{dt}$, $v_z = \frac{dz}{dt}$ and

$$|\vec{v}| = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

When position of a particle is plotted on a coordinate system, $\vec{v}$ is always tangent to the curve representing the path of the particle.

43. **Acceleration vector.** If the velocity of an object changes from $\vec{v}$ to $\vec{v}'$ in time Δt , then its average acceleration is given by

$$\vec{a} = \frac{\vec{v}' - \vec{v}}{\Delta t} = \frac{\Delta\vec{v}}{\Delta t}$$

The acceleration $\vec{a}$ at any time t is the limiting value of $\vec{a}$ as Δt tends to zero. So

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$

In component form, we have

$$\vec{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$$

where, $a_x = \frac{dv_x}{dt}$, $a_y = \frac{dv_y}{dt}$, $a_z = \frac{dv_z}{dt}$

and $|\vec{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}$

44. **Equations of motion in vector form.** For motion with constant acceleration,

$$(i) \vec{v} = \vec{v}_0 + \vec{a}t$$

$$(ii) \vec{r} = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2} \vec{a} t^2$$

$$(iii) v^2 - v_0^2 = 2\vec{a} \cdot (\vec{r} - \vec{r}_0)$$

The motion in a plane with uniform acceleration can be treated as the superposition of two separate simultaneous one-dimensional motions along two perpendicular directions.

45. **Relative velocity.** The relative velocity of an object A with respect to object B, when both are in motion, is the rate of change of position of object A with respect to object B.

- (i) If the two objects are moving with velocities $\vec{v}_A$ and $\vec{v}_B$ with respect to the ground, then

Relative velocity of A w.r.t. B,

$$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$$

Relative velocity of B w.r.t. A,

$$\vec{v}_{BA} = \vec{v}_B - \vec{v}_A$$

- (ii) If the two objects A and B are moving with velocities $\vec{v}_A$ and $\vec{v}_B$ inclined at an angle θ , then magnitude of the relative velocity of A w.r.t. B is given by

$$\begin{aligned} v_{AB} &= \sqrt{|\vec{v}_A|^2 + |\vec{v}_B|^2 + 2|\vec{v}_A||\vec{v}_B|\cos(180^\circ - \theta)} \\ &= \sqrt{v_A^2 + v_B^2 - 2v_A v_B \cos \theta} \end{aligned}$$

If the velocity $\vec{v}_{AB}$ makes angle β with the velocity $\vec{v}_A$, then

$$\begin{aligned} \tan \beta &= \frac{|\vec{v}_B| \sin(180^\circ - \theta)}{|\vec{v}_A| + |\vec{v}_B| \cos(180^\circ - \theta)} \\ &= \frac{v_B \sin \theta}{v_A - v_B \cos \theta} \end{aligned}$$

46. **Projectile motion.** Any body projected into space, such that it moves under the effect of gravity alone is called a projectile. The path followed by a projectile is called its trajectory which is always a parabola. A projectile executes two independent motions simultaneously:

- (i) uniform horizontal motion and
(ii) uniform accelerated downward motion.

47. **Projectile fired horizontally.** Suppose a body is projected horizontally with velocity u from a height h above the ground. Let it reach the point (x, y) after time t .

Then

(i) Position of the projectile after time t :

$$x = ut, \quad y = \frac{1}{2}gt^2$$

(ii) Equation of trajectory : $y = \frac{g}{2u^2} \cdot x^2$

(iii) Velocity after time t :

$$v = \sqrt{u^2 + g^2 t^2}; \quad \beta = \tan^{-1} \frac{gt}{u}$$

(iv) Time of flight, $T = \sqrt{\frac{2h}{g}}$

(v) Horizontal range, $R = u \times T = u \sqrt{\frac{2h}{g}}$

48. Projectile fired at an angle with the horizontal.

Suppose a projectile is fired with velocity u at an angle θ with the horizontal. Let it reach the point (x, y) after time t . Then

(i) Components of initial velocity :

$$u_x = u \cos \theta, \quad u_y = u \sin \theta$$

(ii) Components of acceleration at any instant :

$$a_x = 0, \quad a_y = -g$$

(iii) Position after time t :

$$x = (u \cos \theta)t, \quad y = (u \sin \theta)t - \frac{1}{2}gt^2$$

(iv) Equation of trajectory :

$$y = x \tan \theta - \frac{g}{2u^2 \cos^2 \theta} \cdot x^2$$

(v) Maximum height, $H = \frac{u^2 \sin^2 \theta}{2g}$

(vi) Time of flight, $T = \frac{2u \sin \theta}{g}$

(vii) Horizontal range, $R = \frac{u^2 \sin 2\theta}{g}$

(viii) Maximum horizontal range is attained at $\theta = 45^\circ$ and its value is

$$R_{\max} = \frac{u^2}{g}$$

(ix) Velocity after time t :

$$v_x = u \cos \theta, \quad v_y = u \sin \theta - gt$$

$$\therefore v = \sqrt{v_x^2 + v_y^2} \quad \text{and} \quad \tan \beta = \frac{v_y}{v_x}$$

(x) The velocity with which the projectile reaches the horizontal plane through the point of projection is same as the velocity of projection.

49. **Uniform circular motion.** When a body moves along a circular path with uniform speed, its motion is said to uniform circular motion.

50. **Angular displacement.** It is the angle swept out by a radius vector in a given time interval.

$$\theta (\text{rad}) = \frac{\text{Arc}}{\text{Radius}} = \frac{s}{r}$$

51. **Angular velocity.** The angle swept out by the radius vector per second is called angular velocity.

$$\omega = \frac{\theta}{t} \quad \text{or} \quad \omega = \frac{\theta_2 - \theta_1}{t_2 - t_1}$$

52. **Time period and frequency.** Time taken for one complete revolution is called time period (T). The number of revolutions completed per second is called frequency

$$\omega = \frac{2\pi}{T} = 2\pi\nu$$

53. **Relationship between v and ω .** It is given by

$$v = r\omega$$

i.e. Linear velocity = Radius $\times$ angular velocity.

54. **Angular acceleration and its relation with linear acceleration.** The rate of change of angular velocity is called angular acceleration. It is given by

$$\alpha = \frac{\omega_2 - \omega_1}{t_2 - t_1}$$

Also, $a = r\alpha$

i.e. Linear acceleration

= Radius $\times$ angular acceleration

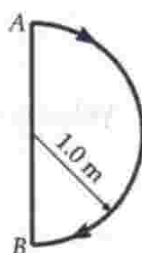
55. **Centripetal acceleration.** A body moving along a circular path is acted upon by an acceleration directed towards the centre along the radius. This acceleration is called centripetal acceleration. It is given by

$$a = \frac{v^2}{r} = r\omega^2$$

IIT Entrance Exam

MULTIPLE CHOICE QUESTIONS WITH ONE CORRECT ANSWER

1. In 1.0 s, a particle goes from point A to point B, moving in a semicircle of radius 1.0 m as shown in the figure.



The magnitude of the average velocity is

- (a) 3.14 m/s (b) 2.0 m/s
(c) 1.0 m/s (d) zero

[IIT 99]

2. A particle P is sliding down a frictionless hemispherical bowl. It passes the point A at $t = 0$. At this instant of time, the horizontal component of its velocity is v . A bead Q of the same mass as P is ejected

Laws of Motion

GLIMPSES

1. **Force.** It may be defined as an agency (a push or pull) which changes or tends to change the state of rest or of uniform motion or the direction of motion of a body. Force is a vector quantity.
2. **Inertia.** It is the inherent property of a material body by virtue of which it cannot change, by itself, its state of rest or of uniform motion in a straight line. Inertia is of three types (i) inertia of rest (ii) inertia of motion and (iii) inertia of direction.
3. **Mass as the measure of inertia.** If a body has more mass, it has more inertia i.e., it is more difficult to change its state of rest or of uniform motion.
4. **Momentum.** It is the quantity of motion in a body. It is equal to the product of mass m and velocity v of the body.

$$\text{Momentum, } p = mv \text{ or } \vec{p} = m\vec{v}$$

Momentum is a vector quantity having the direction of velocity $\vec{v}$. Its SI unit is kg ms^{-1} and CGS unit is g cms^{-1} .

5. **Newton's first law of motion.** It states that every body continues in its state of rest or of uniform motion along a straight line, unless an external force is applied to change that state. This law defines force.
6. **Newton's second law of motion.** It states that the rate of change of momentum of a body is directly proportional to the applied force and the change in momentum takes place in the direction of the applied force. This law gives a measure of the force.

Mathematically,

$$F = \frac{dp}{dt} = \frac{d}{dt}(mv) = m \frac{dv}{dt} = ma = m \left(\frac{v - u}{t} \right)$$

- (i) When m is constant and v changes,

$$F = m \frac{dv}{dt}$$

- (ii) When v is constant and m changes,

$$F = v \frac{dm}{dt}$$

In component form, Newton's second law may be expressed as

$$F_x = \frac{dp_x}{dt} ; F_y = \frac{dp_y}{dt} ; F_z = \frac{dp_z}{dt}$$

7. **Newton's third law of motion.** It states that to every action, there is an equal and opposite reaction. Mathematically,

$$\vec{F}_{BA} = -\vec{F}_{AB}$$

Forces of action and reaction never cancel out because they act on different bodies.

8. **Absolute units of force.** The SI unit of force is *newton* (N) and CGS unit is *dyne* (dyn).

$$1 \text{ newton (N)} = 1 \text{ kg} \times 1 \text{ ms}^{-2} = 1 \text{ kg ms}^{-2}$$

$$1 \text{ dyne (dyn)} = 1 \text{ g} \times 1 \text{ cms}^{-2} = 1 \text{ g cms}^{-2}$$

$$1 \text{ N} = 10^5 \text{ dyne.}$$

9. **Gravitational units of force.** The SI unit of force is *kilogram weight* (kg wt) or *kilogram force* (kg f) and the CGS unit is *gram weight* (g wt) or *gram force* (g f).

$$1 \text{ kg wt} = 1 \text{ kg f} = 1 \text{ kg} \times 9.8 \text{ ms}^{-2} = 9.8 \text{ N}$$

$$1 \text{ g wt} = 1 \text{ g f} = 1 \text{ g} \times 980 \text{ cms}^{-2} = 980 \text{ dyn}$$

10. **Impulse of a force.** Impulse is the total effect of a large force which acts for a short time to produce a finite change in momentum. It is defined as the product of the force and the time for which it acts and equal to the total change in momentum.

$$\text{Impulse} = \text{Force} \times \text{time duration}$$

$$= \text{Total change in momentum.}$$

Impulse is a vector quantity, denoted by $\vec{J}$.

$$\vec{J} = \int_{t_1}^{t_2} \vec{F} \cdot dt$$

= Area under the force-time ($F-t$) graph.

or $\vec{J} = \vec{F}_{av} \times t = \vec{p}_2 - \vec{p}_1 = m(\vec{v} - \vec{u})$

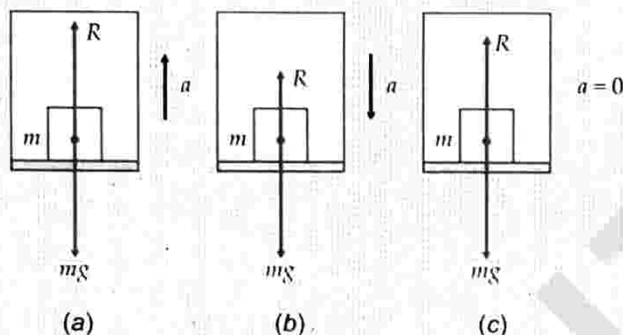
The SI unit of impulse is Ns or kg ms^{-1} and CGS unit is dyn s or g cms^{-1} .

11. Apparent weight of body in a lift.

- (i) When a lift moves upwards with uniform acceleration a , apparent weight of a body in the lift increases.

From Fig. (a),

$$R - mg = ma \quad \text{or} \quad R = m(g + a)$$



- (ii) When a lift moves downwards with acceleration a , the apparent weight of a body in the lift decreases. From Fig. (b),

$$mg - R = ma \quad \text{or} \quad R = m(g - a).$$

- (iii) When a lift is at rest or moves with uniform velocity, $a = 0$, the apparent weight of the body is equal to its true weight.

From Fig. (c),

$$R = mg$$

- (iv) When a lift falls freely, ($a = g$) the apparent weight of a body in the lift becomes zero.

$$R = m(g - g) = 0.$$

12. Law of conservation of linear momentum.

- (i) In the absence of any external force, vector sum of the linear momenta of a system of particles remains constant.

$$\vec{p} = \vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \dots + \vec{p}_n = \text{constant}$$

or $m_1 \vec{v}_1 + m_2 \vec{v}_2 + m_3 \vec{v}_3 + \dots + m_n \vec{v}_n = \text{constant}$

- (ii) When two bodies collide,

Total momentum before collision

= Total momentum after collision

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

- (iii) When a bullet of mass m is fired with velocity v from a gun of mass M , the gun recoils with velocity V .

Momentum of gun = - Momentum of bullet

$$MV = -mv$$

Recoil velocity of gun,

$$V = -\frac{mv}{M}$$

- 13. Rocket propulsion.** It is an example of momentum conservation in which the large backward momentum of the ejected gases imparts an equal forward momentum to the rocket. Due to the decrease in mass of the rocket-fuel system, the acceleration of the rocket keeps on increasing. Let

u = velocity of exhaust gases

v_0, v = initial velocity and velocity of the rocket at any instant t

m_0, m, m_c = initial mass, mass of the rocket at any instant t and mass of empty rocket.

$\frac{dm}{dt}$ = rate of ejection of fuel.

Thrust on rocket : $F = -u \frac{dm}{dt}$

Acceleration of rocket : $a = \left[\frac{u}{m_0 - t \frac{dm}{dt}} \right] \frac{dm}{dt}$

Velocity of rocket : $v = v_0 + u \log_e \frac{m_0}{m}$

Burnt-out speed of rocket : $v_b = v_0 + u \log_e \frac{m_0}{m_c}$

- 14. Concurrent forces.** The forces acting at the same point of a body are called concurrent forces.

- 15. Equilibrium of concurrent forces.** A number of concurrent forces acting on a body are said to be in equilibrium if their vector sum is zero or if these forces can be completely represented by the sides of a closed polygon taken in the same order.

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots + \vec{F}_n = 0$$

- 16. Lami's theorem.** It states that if three forces acting on a particle keep it in equilibrium, then each force is proportional to the sine of the angle between other two forces. If α, β, γ be the angles between $\vec{F}_2$ and $\vec{F}_3$; $\vec{F}_3$ and $\vec{F}_1$; $\vec{F}_1$ and $\vec{F}_2$ respectively, then according to Lami's theorem :

$$\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$$

- 17. Free-body diagram.** A diagram for each body of the system showing all the forces exerted on the body by the remaining parts of the system is called free-body diagram.

18. **Motion of connected bodies.** Suppose two bodies of masses M and m ($M > m$) are tied at the ends of an inextensible string passing over a frictionless pulley. Then

$$\text{Acceleration of the masses, } a = \frac{M - m}{M + m} \cdot g$$

$$\text{Tension in the string, } T = \frac{2Mm}{M + m} \cdot g$$

Clearly, $a < g$.

19. **Friction.** Whenever a body moves or tends to move over the surface of another body, a force comes into play which acts parallel to the surface of contact and opposes the relative motion. This opposing force is called friction.
20. **Static friction.** The force of friction which comes into play between two bodies before one body actually starts moving over the other is called static friction (f_s). Static friction is a self-adjusting force.
21. **Limiting friction.** The maximum force of static friction which comes into play when a body just starts moving over the surface of another body is called limiting friction ($f_s^{\max}$).
22. **Kinetic friction.** The force of friction which comes into play when a body is in a steady motion over the surface of another body is called kinetic or dynamic friction (f_k). Kinetic friction is less than limiting friction.
23. **Laws of limiting friction :**
- The force of limiting friction depends upon the nature of the two surfaces in contact and their state of roughness.
 - The force of limiting friction acts tangential to the two surfaces in contact and in a direction opposite to that of the applied force.
 - The force of limiting friction between any two surfaces is independent of the shape or area of the surfaces in contact so long as the normal reaction remains the same.
 - The force of limiting friction between two given surfaces is directly proportional to the normal reaction between the two surfaces.

$$f \propto R \quad \text{or} \quad f = \mu_s R$$

where the constant of proportionality μ_s is called the coefficient of limiting friction.

24. **Coefficient of limiting friction.** It is the ratio of limiting friction to the normal reaction.

$$\mu_s = \frac{f_s^{\max}}{R} = \frac{\text{Limiting friction}}{\text{Normal reaction}}$$

25. **Coefficient of kinetic friction.** It is the ratio of kinetic friction (f_k) to the normal reaction.

$$\mu_k = \frac{f_k}{R} = \frac{\text{Kinetic friction}}{\text{Normal reaction}}$$

$$\text{As } f_k < f_s^{\max} \quad \text{or} \quad \mu_k R < \mu_s R \quad \therefore \mu_k < \mu_s$$

26. **Angle of friction.** It is the angle which the resultant of the limiting friction and the normal reaction makes with the normal reaction. If θ is the angle of friction, then $\tan \theta = \mu_s$.
27. **Angle of repose.** It is the minimum angle that an inclined plane makes with the horizontal when a body placed on it just begins to slide down. If ϕ is the angle of repose, then $\tan \phi = \mu_s$.

28. **Motion along a rough horizontal surface.** If a body of mass m is moved over a rough horizontal surface through distance s , then

$$\text{Force of friction, } f = \mu R = \mu mg$$

$$\text{Retardation produced, } a = \frac{f}{m} = \mu g$$

$$\text{Work done against friction, } W = f \times s = \mu mg s$$

$$\text{Power} = f \times v = \mu mg v$$

29. **Motion along a rough inclined plane.**

- (i) When a body moves down an inclined plane with uniform velocity ($a = 0$), net downward force needed is

$$F = mg \sin \theta - f = mg (\sin \theta - \mu \cos \theta)$$

$$\text{Work done, } W = F_s = mg (\sin \theta - \mu \cos \theta) s$$

- (ii) When a body moves up an inclined plane with uniform velocity ($a = 0$), net upward force needed is

$$F = mg \sin \theta + f = mg (\sin \theta + \mu \cos \theta)$$

$$W = F_s = mg (\sin \theta + \mu \cos \theta) s$$

- (iii) When a body moves up an inclined plane with acceleration a , net upward force needed is

$$F = ma + mg \sin \theta + f$$

$$= m(a + g \sin \theta + \mu g \cos \theta)$$

$$W = m(a + g \sin \theta + \mu g \cos \theta) s$$

30. **Acceleration of body sliding down a rough inclined plane.** When the angle of inclination is greater than the angle of repose, the acceleration produced is

$$a = g (\sin \theta - \mu_k \cos \theta)$$

31. **Centripetal force.** It is the force required to make a body move along a circular path with a uniform speed. It always acts along the radius and towards the centre of the circular path. The centripetal force required to move a body of mass m along a circular path of radius r with speed v is given by

$$F = \frac{mv^2}{r} = m\omega^2 r = mr(2\pi v)^2 = mr \left(\frac{2\pi}{T} \right)^2$$

32. **Centrifugal force.** It is a fictitious force acting radially outwards on a particle moving in a circle and is equal in magnitude to the centripetal force.

33. **A vehicle taking circular turn on a level road.** If μ is the coefficient of friction between tyres and road, then the maximum velocity with which the vehicle can safely take a circular turn of radius r is given by

$$v = \sqrt{\mu r g}$$

34. **Banking of tracks (roads).** The maximum angle with which a vehicle (in the absence of friction) can negotiate a circular turn of radius r and banked at an angle θ is given by

$$v = \sqrt{r g \tan \theta}$$

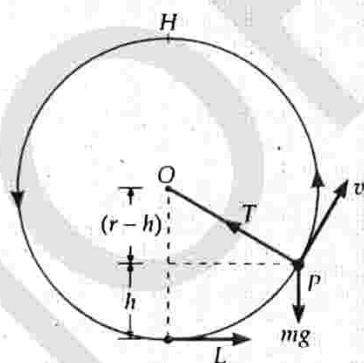
When the frictional forces are also taken into account, the maximum safe velocity is given by

$$v = \sqrt{r g \left(\frac{\mu + \tan \theta}{1 - \mu \tan \theta} \right)}$$

35. **Bending of a cyclist.** In order to take a circular turn of radius r with speed v , the cyclist should bend himself through an angle θ from the vertical such that

$$\tan \theta = \frac{v^2}{r g}$$

36. **Motion in a vertical circle.** As shown in the figure, consider a body of mass m tied at the end of a string and rotating in a vertical circle of radius r .



Then

(i) Velocity at any point P at a height h from the lowest point L ,

$$v = \sqrt{u^2 - 2gh}$$

(ii) Tension in the string at any point P ,

$$T = \frac{m}{r} (u^2 - 3gh + gr)$$

(iii) Tension at the lowest point ($h = 0$),

$$T_L = \frac{m}{r} (u^2 + gr)$$

(iv) Tension at the highest point H ($h = 2r$),

$$T_H = \frac{m}{r} (u^2 - 5gr)$$

(v) Difference in tensions at the highest and lowest points,

$$T_L - T_H = 6mg$$

(vi) Minimum velocity at the lowest point L for looping the loop,

$$v_L = \sqrt{5gr}$$

(vii) Velocity at the highest point for looping the loop,

$$v_H = \sqrt{gr}$$

37. **Inertial frame of reference.** An inertial frame of reference is one in which Newton's first law of motion holds good. All frames moving with uniform motion relative to an inertial frame are also inertial.

38. **Non-inertial frame of reference.** A frame of reference which is accelerating with respect to an inertial frame of reference is called a non-inertial frame of reference. In a non-inertial frame, the second law of motion $\vec{F} = m\vec{a}$ is not valid. Instead, it takes the form

$$\vec{F} - m\vec{\alpha} = m\vec{a} \quad \text{or} \quad \vec{F} + \vec{F}_p = m\vec{a}$$

where $\vec{\alpha}$ is the acceleration of the non-inertial frame relative to any inertial frame and $\vec{a}$ is the acceleration of the body relative to the non-inertial frame. The force $\vec{F}_p = -m\vec{\alpha}$ is an example of pseudo force.

IIT Entrance Exam

MULTIPLE CHOICE QUESTIONS WITH ONE CORRECT ANSWER

1. A ship of mass 3×10^7 kg initially at rest is pulled by a force of 5×10^4 N through a distance of 3 m.

Assuming that the resistance due to water is negligible, the speed of the ship is

- (a) 1.5 m/s (b) 60 m/s
(c) 0.1 m/s (d) 5 m/s

[IIT 80]

Competition Section

Work, Energy and Power

GLIMPSES

- Work.** Work is said to be done wherever a force acts on a body and the body moves through some distance in the direction of the force. If the force $\vec{F}$ makes angle θ with the direction of displacement $\vec{s}$, then the work done is

$$W = \vec{F} \cdot \vec{s} = Fs \cos \theta$$

Work done is a scalar quantity. It can be positive or negative depending on θ is acute or obtuse. The work done by friction or viscous force on a moving body is negative.

Work done is a scalar quantity.

(i) If $\theta = 0^\circ$, $W = Fs$ i.e., work done is maximum.

(ii) If $\theta = 90^\circ$, $W = 0$ i.e., work done is zero.

- Work done against a variable force.** The work done by a variable force $\vec{F}$ in changing the displacement from $\vec{s}_1$ to $\vec{s}_2$ is

$$W = \int_{s_1}^{s_2} \vec{F} \cdot d\vec{s}$$

= Area under the force-displacement curve.

- Units of work.** (i) The SI unit of work is joule. One joule of work is said to be done when a force of one newton displaces a body through a distance of one metre in its own direction.

$$1 \text{ joule} = 1 \text{ newton} \times 1 \text{ metre} \text{ or } 1 \text{ J} = 1 \text{ Nm}$$

(ii) The CGS unit is erg. One erg of work is said to be done if a force of one dyne displaces a body through a distance of one centimetre in its own direction.

$$1 \text{ erg} = 1 \text{ dyne} \times 1 \text{ cm}$$

$$1 \text{ joule} = 10^7 \text{ erg}$$

- Work done in terms of rectangular components.** If

$$\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$$

$$\text{and } \vec{s} = s_x \hat{i} + s_y \hat{j} + s_z \hat{k}$$

$$\text{then } W = F_x s_x + F_y s_y + F_z s_z$$

- Newton's third law and work done.** For two bodies, the sum of the mutual forces exerted between the two bodies is zero from Newton's third law,

$$\vec{F}_{12} + \vec{F}_{21} = \vec{0}$$

But the sum of the work done by the two forces need not always cancel, i.e.,

$$W_{12} + W_{21} \neq 0.$$

However, it may sometimes be true.

- Energy.** It is defined as the ability of a body to do work. It is measured by the amount of work that a body can do. It is a scalar quantity. Like work, SI unit of energy is joule and the CGS unit is erg. The unit of energy used at the atomic level is electron volt (eV).

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

$$1 \text{ keV} = 10^3 \text{ eV} = 1.6 \times 10^{-16} \text{ J}$$

$$1 \text{ MeV} = 10^6 \text{ eV} = 1.6 \times 10^{-13} \text{ J}$$

- Kinetic energy.** It is the energy possessed by a body by virtue of its motion. The K.E. of a body of mass m moving with speed v is

$$K = \frac{1}{2} mv^2 = \frac{p^2}{2m}$$

- Work-energy theorem.** The work done by the net force acting on a body is equal to the change in kinetic energy of the body.

$$W = \text{Change in K.E.} = \frac{1}{2} mv^2 - \frac{1}{2} mu^2$$

The WE theorem may be regarded as the scalar form of Newton's second law of motion.

9. **Potential energy.** It is the energy possessed by a body by virtue of its position (in a field) or configuration (shape or size).

For a conservative force in one dimension, the potential energy functions $U(x)$ may be defined as

$$F(x) = - \frac{dU(x)}{dx}$$

or
$$\Delta U = U_f - U_i = - \int_{x_i}^{x_f} F(x) dx$$

10. **Gravitational potential energy.** It is the energy possessed by a body by virtue of its position above the surface of the earth. The gravitational P.E. of a body of mass m at a height h above the earth's surface is $U = mgh$

11. **Potential energy of a spring.** According to Hooke's law, when a spring is stretched through distance x , the restoring force set up in the spring due to its elasticity is such that

$$F \propto x \quad \text{or} \quad F = -kx$$

where k is the force constant or spring constant of the spring. It is the restoring force set up in the spring per unit extension. Its unit is Nm^{-1} . The work done in stretching the spring through distance x will be

$$W = \int_0^x kx dx = \frac{1}{2} kx^2$$

This work done is stored as potential energy U of the spring.

$$\therefore U = \frac{1}{2} kx^2$$

12. **Conservative force.** A force is conservative (i) if the work done by the force in displacing a particle from one point to another is independent of the path followed by the particle and (ii) if the work done by the force in moving a particle around any closed path is zero. Gravitational force, electrostatic force and elastic force of a spring are all conservative forces.
13. **Non-conservative force.** If the amount of work done in moving an object against a force from one point to another depends on the path along which the body moves, then such a force is called a non-conservative force. Forces of friction and viscosity are non-conservative forces.
14. **Mass-energy equivalence.** According to Einstein, mass can be converted into energy and energy into mass. A mass m is equivalent to energy E given by

$$E = mc^2$$

where c is the speed of light in vacuum and $c = 3 \times 10^8 \text{ ms}^{-1}$.

15. **Power.** It is the rate of doing work. It is a scalar quantity.

$$\text{Power} = \frac{\text{Work}}{\text{Time}} \quad \text{or} \quad P = \frac{W}{t}$$

Instantaneous power is given by

$$P = \frac{dW}{dt} = \frac{d}{dt} (\vec{F} \cdot \vec{s}) = \vec{F} \cdot \frac{d\vec{s}}{dt} = \vec{F} \cdot \vec{v} = Fv \cos \theta$$

If $\theta = 0^\circ$, then $P = Fv$.

16. **Units of power.** The SI unit of power is watt (W).

$$1 \text{ watt} = \frac{1 \text{ joule}}{1 \text{ second}} \quad \text{or} \quad 1 \text{ W} = 1 \text{ J s}^{-1}$$

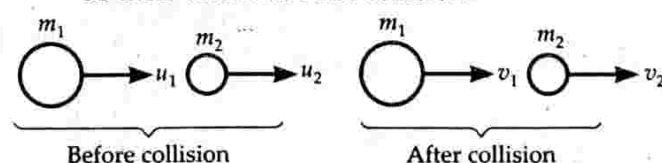
The larger units of power are kilowatt (kW) and horse power (hp).

$$1 \text{ kW} = 1000 \text{ W} \quad \text{and} \quad 1 \text{ hp} = 746 \text{ W}$$

17. **Kilowatt hour.** It is the commercial unit of electrical energy. One kilowatt hour is the electrical energy consumed by an appliance of 1000 watt in 1 hour.

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

18. **Law of conservation of energy.** It states that energy can neither be created nor destroyed but can only be transformed from one form to another.
19. **Collision.** A collision between two bodies is said to occur if either they physically collide against each other or the path of the motion of one body is influenced by the other.
20. **Elastic collision.** If there is no loss of kinetic energy during a collision, it is called an elastic collision.
21. **Inelastic collision.** If there is a loss of kinetic energy during a collision, it is called an inelastic collision. Linear momentum is conserved both in elastic and inelastic collisions.
22. **Perfectly inelastic collision.** It is the collision in which two bodies stick together after the collision.
23. **Head-on collision or one-dimensional collision.** It is a collision in which the colliding bodies move along the same straight line path before and after the collision.
24. **Oblique collision.** If the two bodies do not move along the same straight line path before and after the collision, the collision is said to be oblique collision.
25. **Velocities in one-dimensional elastic collision.** Suppose two bodies of masses m_1 and m_2 moving with velocities u_1 and u_2 ($u_1 > u_2$) in the same direction suffer head-on elastic collision. Let v_1 and v_2 be their velocities after collision.



By the law of conservation of momentum,

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

As K.E. is conserved in an elastic collision, so

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

By solving the above two equations, it can be shown that in an elastic collision,

Velocity of approach = Velocity of separation

$$\text{or } u_1 - u_2 = v_2 - v_1$$

$$\text{Also, } v_1 = \frac{m_1 - m_2}{m_1 + m_2} u_1 + \frac{2m_2}{m_1 + m_2} u_2$$

$$\text{and } v_2 = \frac{2m_1}{m_1 + m_2} u_1 + \frac{m_2 - m_1}{m_1 + m_2} u_2$$

26. **Coefficient of restitution.** The coefficient of restitution for a collision between two bodies is the ratio of the magnitude of their relative velocity after collision to the magnitude of their relative velocity before the collision.

$$e = -\frac{v_1 - v_2}{u_1 - u_2} = \frac{|v_1 - v_2|}{|u_1 - u_2|}$$

For a perfectly elastic collision, $e = 1$ and for a perfectly inelastic collision, $e = 0$. Hence $0 \leq e \leq 1$.

For a ball rebounding from a floor,

$$e = \frac{v}{u}$$

where u and v are the magnitudes of the velocities of the ball before and after the collision respectively.

IIT Entrance Exam

MULTIPLE CHOICE QUESTIONS WITH ONE CORRECT ANSWER

1. Two masses of 1 g and 4 g are moving with equal kinetic energy. The ratio of the magnitudes of their momenta is

- (a) 4 : 1 (b) $\sqrt{2}$: 1
(c) 1 : 2 (d) 1 : 16

[IIT 80]

2. If a machine is lubricated with oil

- (a) the mechanical advantage of the machine increases
(b) the mechanical efficiency of the machine increases
(c) both its mechanical advantage and efficiency increase
(d) its efficiency increases, but its mechanical advantage decreases.

[IIT 80]

3. A spring of force-constant k is cut into two pieces such that one piece is double the length of the other. Then the long piece will have a force-constant of

- (a) $(2/3)k$ (b) $(3/2)k$
(c) $3k$ (d) $6k$

[IIT 99]

4. An ideal spring with spring constant k is hung from the ceiling and a block of mass M is attached to its lower end. The mass is released with the spring initially unstretched. Then the maximum extension in the spring is

- (a) $\frac{4Mg}{k}$ (b) $\frac{2Mg}{k}$
(c) $\frac{Mg}{k}$ (d) $\frac{Mg}{2k}$

[IIT 02]

5. A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration a_c is varying with time t as $a_c = k^2 r t^2$, where k is a constant. The power delivered to the particle by the force acting on it is

- (a) $2\pi m k^2 r^2 t$ (b) $m k^2 r^2 t$
(c) $\frac{(m k^4 r^2 t^5)}{3}$ (d) zero

[IIT 94]

6. A wind-powered generator converts wind energy into electrical energy. Assume that the generator converts a fixed fraction of the wind energy intercepted by its blades into electrical energy. For wind speed v , the electrical power output will be proportional to

- (a) v (b) v^2
(c) v^3 (d) v^4

[IIT 2K]

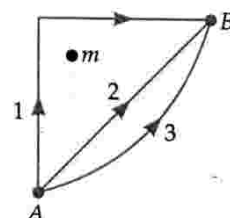
7. If W_1 , W_2 and W_3 represent the work done in moving a particle from A to B along three different paths 1, 2 and 3 respectively (as shown) in the gravitational field of a point mass m , find the correct relation between W_1 , W_2 and W_3 .

- (a) $W_1 > W_2 > W_3$ (b) $W_1 = W_2 = W_3$
(c) $W_1 < W_2 < W_3$ (d) $W_2 > W_1 > W_3$

[IIT 03]

8. A particle is acted by a force $F = kx$, where k is a +ve constant. Its potential energy at $x = 0$ is zero. Which curve correctly represents the variation of potential energy of the block with respect to x ?

[IIT 04]



Systems of Particles & Rotational Motion

GLIMPSES

1. **Particle.** A particle is an object whose mass is finite but whose size and internal structure can be neglected.
2. **System.** A system is a collection of a very large number of particles which mutually interact with one another.
3. **Centre of mass.** It is the point at which entire mass of a system may be supposed to be concentrated. The nature of motion of the system remains unaffected when all the forces acting on the system are applied directly on the centre of mass of the system.

If $\vec{r}_1$ and $\vec{r}_2$ are the position vectors of two particles of masses m_1 and m_2 , then the position vector of their centre of mass is given by

$$\vec{R}_{CM} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

For a system of N -particles, the centre of mass is given by

$$\begin{aligned}\vec{R}_{CM} &= \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + \dots + m_N \vec{r}_N}{m_1 + m_2 + \dots + m_N} \\ &= \frac{\sum_{i=1}^N m_i \vec{r}_i}{M}\end{aligned}$$

For a continuous mass distribution, the centre of mass is given by

$$\vec{R}_{CM} = \frac{1}{M} \int \vec{r} dm$$

where dm is the mass of small element located at position $\vec{r}$

The velocity of the centre of mass of a two-particle system is given by

$$\vec{v}_{CM} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2}$$

4. Properties of centre of mass :

- (i) The location of the centre of mass is the weighted average of the locations of the particles of the system.
- (ii) Centre of mass moves as if the external force acts on the entire mass concentrated at this point.
- (iii) In the absence of any external force, the centre of mass moves with a constant velocity.
- (iv) For bodies of normal dimensions, centre of mass and centre of gravity coincide.

5. **Rigid body.** A rigid body is one whose constituent particles retain their relative positions even when they move under the action of an external force.

6. **Rotational motion of a rigid body.** A body is said to possess rotational motion if all its particles move along circles in parallel planes. The centres of these circles lie on a fixed line perpendicular to the parallel planes and is called the axis of rotation.

7. **Equations of rotational motion.** For a body moving in circle, let ω_0 be the initial angular velocity, ω the final angular velocity, α the angular acceleration and θ be the angular displacement after the time t . Then equations of rotational motion can be written as

$$(i) \omega = \omega_0 + \alpha t \quad (ii) \theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$(iii) \omega^2 - \omega_0^2 = 2\alpha\theta$$

8. **Torque ($\vec{\tau}$).** The turning effect of a force about the axis of rotation is called moment of force or torque due to the force.

Torque = Force $\times$ its perpendicular distance
from the axis of rotation

or $\tau = \text{Force} \times \text{moment arm} = Fd$

If a force $\vec{F}$ acts at a point whose position vector is $\vec{r}$ with respect to the axis of rotation, then torque is

$$\vec{\tau} = \vec{r} \times \vec{F}$$

(i) In Cartesian coordinates : $\tau_z = x F_y - y F_x$

(ii) In polar coordinates : $\tau = rF \sin \theta$

where θ is the angle between force vector $\vec{F}$ and position vector $\vec{r}$.

SI Unit of torque is Nm.

9. **Principle of moments of rotational equilibrium.** When a body is in rotational equilibrium, the sum of the clockwise moments about any point is equal to the sum of the anticlockwise moments about that point. Or, the algebraic sum of moments about any point is zero. In rotational equilibrium,
Clockwise moment = Anticlockwise moment

or $F_1 \times d_1 = F_2 \times d_2$

or Load $\times$ load arm = Effort $\times$ effort arm

This is sometimes called the lever principle.

10. **Couple.** A pair of equal and opposite forces acting on a body along two different lines of action constitute a couple. In a couple, total external force is zero but torque is non-zero. So a couple has a turning effect but cannot produce translational motion.

11. **Moment of a couple.** The moment of a couple is equal to the product of either of the forces and the perpendicular distance, called the arm of the couple, between their lines of action. It is independent of the choice of the fulcrum or the point of rotation.

Moment of couple = Force $\times$ arm of the couple

or $\tau = Fd$

A couple can only be balanced by an equal and opposite couple.

12. **Work done by a torque and power of a torque.** If a torque τ applied on a body rotates it through an angle $\Delta\theta$, the work done by the torque is

$$\Delta W = \tau \Delta\theta$$

or Work done = Torque $\times$ Angular displacement

Power of a torque is given by

$$P = \frac{\Delta W}{\Delta t} = \frac{\tau \Delta\theta}{\Delta t} = \tau \omega$$

i.e., Power of a torque = Torque $\times$ Angular velocity

13. **Angular momentum ($\vec{L}$).** It is the moment of linear momentum of a particle about the axis of rotation.

Angular momentum = Linear momentum

$\times$ its perpendicular distance
from the axis of rotation.

or $L = p d$

If $\vec{p}$ is the linear momentum of a particle and $\vec{r}$ its position vector, then its angular momentum is

$$\vec{L} = \vec{r} \times \vec{p}$$

(i) In Cartesian coordinates : $L_z = x p_y - y p_x$

(ii) In polar coordinates : $L = r p \sin \theta$

where θ is the angle between the linear momentum vector $\vec{p}$ and the position vector $\vec{r}$.

If a particle of mass m moves with uniform speed v along a circle of radius r , then its angular momentum is $L = mvr$

SI unit of angular momentum is $\text{kgm}^2 \text{s}^{-1}$.

The linear momentum $\vec{p}$ and velocity vector $\vec{v}$ are always parallel to each other. But the angular momentum $\vec{L}$ and the angular velocity $\vec{\omega}$ are not necessarily parallel vectors.

14. **Geometrical meaning of angular momentum.** Geometrically, the angular momentum of a particle is equal to twice the product of its mass and areal velocity. Equivalently, the areal velocity of a particle is just half its angular momentum per unit mass.

$$\vec{L} = 2m \frac{\Delta \vec{A}}{\Delta t} \quad \text{or} \quad \frac{\Delta \vec{A}}{\Delta t} = \frac{\vec{L}}{2m}$$

15. **Relation between torque and angular momentum.** The rate of change of angular momentum of a system of particles about a fixed point is equal to the total external torque acting on the system about that point.

$$\vec{\tau}_{\text{tot}} = \frac{d\vec{L}}{dt}$$

16. **Equilibrium of rigid bodies.** A rigid body is said to be in equilibrium if both the linear momentum and angular momentum of the rigid body remain constant with time. It must possess the following two equilibria simultaneously :

(i) **Translational equilibrium.** The resultant of all the external forces acting on the body must be zero.

$$\sum \vec{F}_i^{\text{ext}} = 0$$

(ii) **Rotational equilibrium.** The resultant of all the torques due to all the forces acting on the body about any point must be zero.

$$\sum \vec{\tau}_i^{\text{ext}} = \sum \vec{r}_i \times \vec{F}_i^{\text{ext}} = 0$$

17. **Moment of inertia.** The moment of inertia of a rigid body about an axis of rotation is the sum of the products of masses of the various particles and squares of their perpendicular distances from the axis of rotation. Mathematically

$$I = m_1 r_1^2 + m_2 r_2^2 + \dots + m_n r_n^2 = \sum_{i=1}^n m_i r_i^2$$

SI unit of moment of inertia = kg m^2 .

The moment of inertia of a body can be regarded as the rotational inertia of a body. It is the rotational analogue of mass in linear motion.

18. **Factors on which the moment of inertia depends.**

The moment of inertia of a body depends on the following factors :

- (i) Mass of the body.
- (ii) Size and shape of the body.
- (iii) Distribution of mass about the axis of rotation.
- (iv) Position and orientation of the axis of rotation with respect to the body.

19. **Radius of gyration.** It may be defined as the distance from the axis of rotation at which if whole mass of the body were supposed to be concentrated, the moment of inertia, would be same as with the actual distribution of mass. The relation between moment of inertia I and radius of gyration K is

$$I = MK^2 \quad \text{or} \quad K = \sqrt{\frac{I}{M}}$$

For a body composed of particles of equal masses,

$$K = \sqrt{\frac{r_1^2 + r_2^2 + \dots + r_n^2}{n}}$$

i.e. radius of gyration is equal to the root mean square distance of the particles from the axis of rotation.

SI unit of radius of gyration = m.

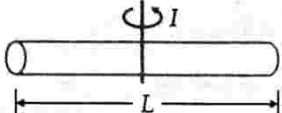
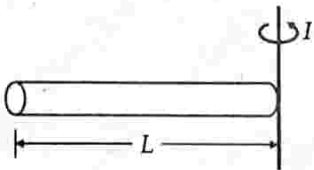
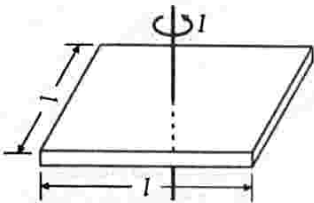
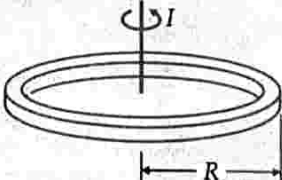
20. **Theorem of perpendicular axes.** It states that the moment of inertia of a plane lamina about an axis perpendicular to its plane is equal to the sum of the moments of inertia of the lamina about any two mutually perpendicular axes in its plane and intersecting each other at the point, where the perpendicular axes pass through the lamina. Mathematically,

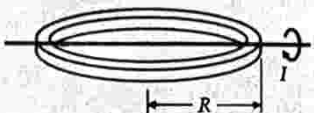
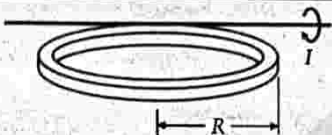
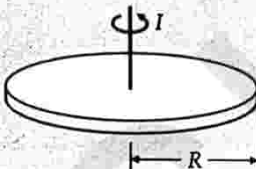
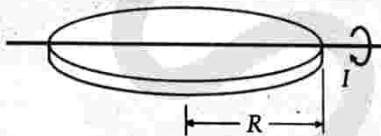
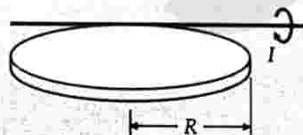
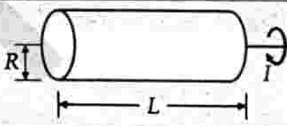
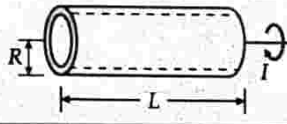
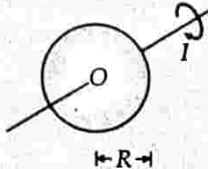
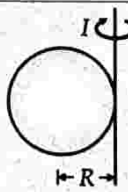
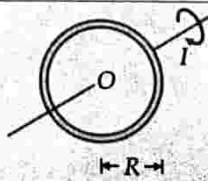
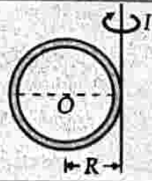
$$I_z = I_x + I_y$$

where X- and Y-axes lie in the plane of the lamina and Z-axis is perpendicular to its plane and passes through the point of intersection of X- and Y-axes.

21. **Theorem of parallel axes.** It states that the moment of inertia of a rigid body about any axis is equal to the moment of inertia of the body about a parallel axis through its centre of mass plus the product of mass of the body and the square of the perpendicular distance between the parallel axes. Mathematically, $I = I_{CM} + Md^2$

Table Moments of Inertia of some Bodies of Regular Shape

Body	Axis	Figure	Moment of Inertia
Thin rod of length L	Passing through centre and perpendicular to the rod		$I = \frac{1}{12} ML^2$
Thin rod of length L	Through its one end and perpendicular to its length		$I = \frac{1}{3} ML^2$
Rectangular lamina of length l and breadth b	Through its centre and perpendicular to its plane		$I = M \left(\frac{l^2 + b^2}{12} \right)$
Circular ring of radius R	Passing through its centre and perpendicular to its plane		$I = MR^2$

Body	Axis	Figure	Moment of Inertia
Circular ring of radius R	Diameter		$I = \frac{1}{2} MR^2$
Circular ring of radius R	Tangent in its plane		$I = \frac{3}{2} MR^2$
Circular disc of radius R	Passing through its centre and perpendicular to its plane		$I = \frac{1}{2} MR^2$
Circular disc of radius R	Diameter		$I = \frac{1}{4} MR^2$
Circular disc of radius R	Tangent in its plane		$I = \frac{5}{4} MR^2$
Right circular solid cylinder of radius R and length L	Symmetry axis		$I = \frac{1}{2} MR^2$
Hollow cylinder of radius R and length L	Symmetry axis		$I = MR^2$
Solid sphere of radius R	Diameter		$I = \frac{2}{5} MR^2$
Solid sphere of radius R	Any tangent		$I = \frac{7}{5} MR^2$
Hollow sphere of radius R	Symmetry axis		$I = \frac{2}{3} MR^2$
Hollow sphere of radius R	Any tangent		$I = \frac{5}{3} MR^2$

22. **Rotational K.E.** If a body of mass M and moment of inertia I rotates about an axis of rotation with an angular velocity ω , then

$$\text{Rotational K.E.} = \frac{1}{2} I \omega^2.$$

23. **Total K.E. of a rolling body.** The centre of mass of a rolling body moves along a straight line, so it possesses translational K.E. in addition to rotational K.E.

$\therefore$ Total K.E. of a rolling body

$$= \text{Translational K.E.} + \text{Rotational K.E.}$$

$$= \frac{1}{2} M v^2 + \frac{1}{2} I \omega^2$$

24. **Relation between M.I. and angular momentum.**

$$\text{Angular momentum} = \text{M.I.} \times \text{Angular velocity}$$

$$\text{i.e., } L = I \omega$$

25. **Relation between M.I. and torque.**

$$\text{Torque} = \text{M.I.} \times \text{Angular acceleration}$$

$$\text{i.e., } \tau = I \alpha.$$

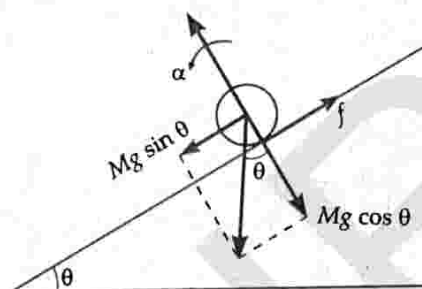
26. **Law of conservation of angular momentum.** If no external torque acts on a system, total angular momentum of the system remains unchanged.

In the absence of any external torque,

$$L = I \omega = \text{constant}$$

$$\text{or } I_1 \omega_1 = I_2 \omega_2 \quad \text{or } I_1 \cdot \frac{2\pi}{T_1} = I_2 \cdot \frac{2\pi}{T_2}$$

27. **Motion of a cylinder rolling down an inclined plane without slipping.** As shown in figure, consider a solid cylinder of mass M and radius R rolling down an inclined plane of inclination θ without slipping.



- (i) Linear acceleration of the cylinder down the inclined plane,

$$a = \frac{g \sin \theta}{1 + \frac{I}{MR^2}} = \frac{2}{3} g \sin \theta$$

- (ii) Force of friction between inclined plane and cylinder,

$$f = \frac{1}{3} Mg \sin \theta$$

- (iii) Condition for rolling of cylinder without slipping is

$$\frac{1}{3} \tan \theta \leq \mu_s$$

where μ_s is the coefficient of static friction.

IIT Entrance Exam

MULTIPLE CHOICE QUESTIONS WITH ONE CORRECT ANSWER

1. Look at the drawing given in the figure which has been drawn with ink of uniform line-thickness. The mass of ink used to draw each of the two inner circles, and each of the two line segments is m . The mass of the ink used to draw the outer circle is $6m$. The coordinates of the centres of the different parts are :

outer circle $(0, 0)$, left inner circle

$(-a, a)$, right inner circle (a, a) , vertical

line $(0, 0)$ and horizontal line $(0, -a)$.

The y -coordinate of the centre of

mass of the ink in this drawing is

(a) $\frac{a}{10}$

(b) $\frac{a}{8}$

(c) $\frac{a}{12}$

(d) $\frac{a}{3}$

[IIT 09]

2. Two particles A and B, initially at rest, move towards each other under mutual force of attraction. At the instant when the speed of A is v and the speed of B is $2v$, the speed of the centre of mass of the system is

(a) $3v$

(b) v

(c) $1.5v$

(d) zero.

[IIT 82]

3. Two blocks of masses 10 kg and 4 kg are connected by a spring of negligible mass and placed on a frictionless horizontal surface. An impulse gives a velocity of 14 m/s to the heavier block in the direction of the lighter block.

The velocity of the centre of mass is

(a) 30 m/s

(b) 20 m/s

(c) 10 m/s

(d) 5 m/s

[IIT 02]

4. An isolated particle of mass m is moving in horizontal plane (x - y) along the x -axis at a certain height above the ground. It suddenly explodes into two fragments of masses $m/4$ and $3m/4$. An instant later, the smaller fragment is at $y = +15$ cm. The larger fragment at this instant is at

(a) $y = -5$ cm

(b) $y = +20$ cm

(c) $y = +5$ cm

(d) $y = -20$ cm.

[IIT 97]

Gravitation

GLIMPSES

1. **Gravitation and gravity.** Gravitation is the force of attraction between any two bodies while gravity refers to the force of attraction between any body and the earth.

2. **Newton's law of gravitation.** It states that every body in this universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

Mathematically, $F = G \frac{m_1 m_2}{r^2}$

where G is the universal gravitational constant. It was derived by Newton on the basis of Kepler's laws of planetary motion.

3. **Universal gravitational constant (G).** It is equal to the force of attraction between two bodies of unit mass each and separated by unit distance.

In SI units, $G = 6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$

In CGS system, $G = 6.67 \times 10^{-8} \text{ dyne cm}^2 \text{ g}^{-2}$

Dimensional formula of G is $[M^{-1}L^3T^{-2}]$.

4. **Properties of gravitational forces.** The gravitational force between two point masses (i) is independent of intervening medium (ii) obeys Newton's third law of motion (iii) has spherical symmetry (iv) is independent of the presence of other bodies (v) obeys principle of superposition (vi) is conservative and central.

5. **Principle of superposition of gravitational forces.** The gravitational force between two masses acts independently and uninfluenced by the presence of other bodies. Hence the resultant force on a particle due to a number of masses is the vector sum of the gravitational forces exerted by the individual masses on the given particle.

$$\vec{F}_R = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots + \vec{F}_n = \sum_{i=1}^n \vec{F}_i$$

6. **Shell theorem.** (i) The gravitational force on a particle placed anywhere inside a spherical shell is zero (ii) If a particle lies outside a spherical shell, the shell attracts the particle as though the mass of the shell were concentrated at the centre of the shell.

7. **Free fall.** The motion of a body under the influence of gravity alone is called a free fall.

8. **Acceleration due to gravity (g).** The acceleration produced in the motion of a body under the effect of gravity is called acceleration due to gravity.

At the surface of the earth, $g = \frac{GM}{R^2}$

where M is the mass and R the radius of the earth. It is a vector quantity. Its SI unit is ms^{-2} .

9. **Mass of the earth.** It is given by

$$M = \frac{gR^2}{G}$$

10. **Mean density of the earth.** If the earth is a sphere of mass M and mean density ρ , then its mass would be

$$M = \text{Volume} \times \text{density} = \frac{4}{3} \pi R^3 \rho$$

Also, $M = \frac{gR^2}{G}$

$$\therefore \frac{4}{3} \pi R^3 \rho = \frac{gR^2}{G}$$

or $\rho = \frac{3g}{4\pi GR}$

11. **Weight of a body.** It is the gravitational force with which a body is attracted towards the centre of the earth.

$$\vec{W} = m \vec{g}$$

Weight of a body is a vector quantity. It is measured in newton, kg wt, etc. The weight of a body varies from place to place because of the variation in the value of g .

12. Variation of acceleration due to gravity.**(i) Effect of altitude.** At a height h ,

$$g_h = g \left(1 + \frac{h}{R} \right)^{-2}$$

and $g_h = g \left(1 - \frac{2h}{R} \right)$, when $h \ll R$.

Clearly, the value of g decreases with the increase in h .

(ii) Effect of depth. At a depth d ,

$$g_d = g \left(1 - \frac{d}{R} \right)$$

Clearly, the value of g decreases with the increase in depth d and becomes zero at the centre of the earth.

(iii) Effect of rotation. If ω is the angular velocity of the earth about its axis of rotation, then at latitude λ ,

$$g_\lambda = g - R\omega^2 \cos^2 \lambda$$

As λ increases, $\cos \lambda$ decreases and g_λ increases. So as we move from equator to pole, the value of g increases.

At equator, $\lambda = 0^\circ$, $\cos \lambda = 1$, hence

$$g_{\text{equa}} = g - R\omega^2$$

At poles, $\lambda = 90^\circ$, $\cos \lambda = 0$

hence $g_{\text{pole}} = g - R\omega^2 \times 0 = g$

Hence g is minimum at equator and maximum at poles.

$$g_{\text{pole}} - g_{\text{equa}} = g - (g - R\omega^2) = R\omega^2$$

(iv) Effect of non-sphericity of the earth. The earth has an equatorial bulge and is flattened at the poles. As $R_e > R_p$, so the value of g is minimum at the equator and maximum at the poles.**13. Gravitational field.** It is the space around a material body in which its gravitational pull can be experienced by other bodies.**14. Intensity of gravitational field.** The intensity or strength of gravitational field at any point in a gravitational field is equal to the force experienced by a unit mass placed at that point. It is a vector quantity directed towards the body producing the gravitational field. It is given by

$$E = \frac{\text{Force}}{\text{Mass}} = \frac{F}{m} = \frac{GM}{r^2}$$

The intensity of gravitational field at a point is equal to the acceleration due to gravity at that point.

SI unit of E is N kg^{-1} and the CGS unit is dyne g^{-1} .

15. Gravitational potential. The gravitational potential at a point in the gravitational field of a body is the amount of work done in bringing a body of unit mass from infinity to that point. It is a scalar quantity.

$$\text{Gravitational potential, } V = \frac{\text{Work done}}{\text{Mass}} = -\frac{GM}{r}$$

SI unit of V is J kg^{-1} and its CGS unit is erg g^{-1} .

16. Gravitational potential energy. The gravitational potential energy of a body may be defined as the energy associated with it due to its position in the gravitational field of another body and is measured by the amount of work done in bringing a body from infinity to a given point in the gravitational field of the other body.

Gravitational P.E. = Gravitational potential $\times$ mass of body

$$U = -\frac{GM}{r} \times m$$

Gravitational intensity (E) and gravitational potential (V) at a point are related as

$$E = -\frac{dV}{dr}$$

17. Escape velocity (v_e). It is the minimum velocity with which a body must be projected vertically upwards in order that it may just escape the gravitational field of the earth.

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}$$

For earth, the value of escape velocity is 11.2 kms^{-1} . It is independent of the mass of body projected.

18. Satellite. It is a heavenly or an artificial body which is revolving continuously in an orbit around a planet.**19. Orbital velocity of a satellite (v_0).** It is the velocity required to put a satellite in its orbit around a planet. The orbital velocity of a satellite revolving around the earth at a height h is given by

$$v_0 = \sqrt{\frac{GM}{R+h}} = \sqrt{\frac{gR^2}{R+h}} = R \sqrt{\frac{g}{R+h}}$$

When a satellite revolves close to the surface of the earth, ($h = 0$),

$$v_0 = \sqrt{gR} \quad \text{and} \quad v_e = \sqrt{2} v_0$$

20. Time period of a satellite (T). It is the time taken by a satellite to go once around the planet

$$T = \frac{\text{Circumference of the orbit}}{\text{Orbital velocity}}$$

$$\begin{aligned}
 &= \frac{2\pi(R+h)}{v_0} = \frac{2\pi(R+h)}{\sqrt{\frac{GM}{R+h}}} \\
 &= 2\pi \sqrt{\frac{(R+h)^3}{GM}} = 2\pi \sqrt{\frac{(R+h)^3}{gR^2}} \\
 &= \sqrt{\frac{3\pi(R+h)^3}{G\rho R^3}}
 \end{aligned}$$

If the satellite revolves just close to the surface of the earth, $h = 0$, then

$$T = 2\pi \sqrt{\frac{R^3}{GM}} = 2\pi \sqrt{\frac{R}{g}} = \sqrt{\frac{3\pi}{G\rho}}$$

21. **Height of a satellite above the earth's surface.** We know that

$$T = 2\pi \sqrt{\frac{(R+h)^3}{gR^2}} \quad \text{or} \quad (R+h)^3 = \frac{T^2 R^2 g}{4\pi^2}$$

$$\text{or} \quad h = \left[\frac{T^2 R^2 g}{4\pi^2} \right]^{1/3} - R$$

22. **Angular momentum of a satellite.** The angular momentum of a satellite of mass m moving with velocity v_0 in an orbit of radius $r (= R + h)$ is given by

$$L = m_0 v r = m \sqrt{\frac{GM}{r}} \cdot r = \sqrt{GMm^2 r}$$

23. **Total energy of a satellite.** For a satellite of mass m moving with velocity v_0 in an orbit of radius r ,

$$\text{Potential energy, } U = -\frac{GMm}{r}$$

$$\text{Kinetic energy, } K = \frac{1}{2} m v_0^2 = \frac{1}{2} m \left(\frac{GM}{r} \right)$$

$$\text{Total energy, } E = K + U$$

$$\text{or} \quad E = \frac{1}{2} m v_0^2 - \frac{GMm}{r} = -\frac{GMm}{2r}$$

$$\text{Clearly, } E = -K = U/2$$

Negative total energy indicates that the satellite is bound to the earth.

24. **Geostationary or synchronous satellite.** It is satellite which revolves around the earth with the same angular speed and in the same direction as the earth rotates about its own axis. Such a satellite should revolve around the earth from west to east in an orbit concentric and coplanar with the equatorial plane of the earth at a height of 36,000 km.

Three geostationary satellites with a mutual angular separation of 120° can be used to communicate between any of two points of the entire earth.

25. **Polar satellite.** A satellite that revolves in a polar orbit is called a polar satellite. Such a satellite passes once over geographical north and south poles during each round trip. A polar orbit has a smaller radius of 500 – 800 km.

26. **Conservation of quantities in motion under gravitational influence.** During the motion of an object under the gravitational influence of another object the following quantities are conserved : (a) Angular momentum (b) Total mechanical energy.

Linear momentum is not conserved. Conservation of angular momentum leads to Kepler's second law of planetary motion.

27. **Kepler's laws of planetary motion**

(i) **Law of orbits.** Every planet moves in an elliptical orbit around the sun, with the sun being at one of the focii.

(ii) **Law of areas.** The radius vector drawn from the sun to the planet sweeps out equal areas in equal intervals of time i.e., the areal velocity of a planet is constant.

(iii) **Law of periods or Harmonic law.** The square of the period of revolution (T) of a planet around the sun is proportional to the cube of the semi-major axis r of the ellipse.

$$T^2 \propto r^3$$

28. **Weightlessness.** A body is said to be in a state of weightlessness when the reaction of the supporting surface is zero or its apparent weight is zero. An astronaut experiences weightlessness in a space satellite. This is not because the gravitational force is small at that location in space. It is because both the astronaut and the satellite are in the state of free fall towards the earth.

29. **Inertial mass.** The mass of a body which measures its inertia and which is given by the ratio of the external force applied on it to the acceleration produced in it along a smooth horizontal surface is called inertial mass.

$$\text{Inertial mass} = \frac{\text{Applied force}}{\text{Acceleration produced}} \quad \text{or} \quad m_i = \frac{F}{a}$$

30. **Gravitational mass.** The mass of a body which determines the gravitational pull due to earth acting upon it is called gravitational mass. On the surface of the earth,

$$F = \frac{GMm_g}{R^2}$$

$$\therefore \text{Gravitational mass, } m_g = \frac{FR^2}{GM}$$

Inertial mass of a body is equivalent to its gravitational mass.

Competition Section

Mechanical Properties of Solids

GLIMPSES

- Interatomic force.** It is the force between the atoms of a molecule. It arises due to the electrostatic interaction between the nuclei of two atoms, their electron clouds and between the nucleus of one atom and the electron cloud of the other atom.
- Intermolecular forces.** It is the force acting between the two molecules of a substance due to electrostatic interaction between their oppositely charged ends. Such forces operate over distances of 10^{-9} m and are weaker than interatomic forces.
- Solids.** A solid is a large accumulation of ($\sim 10^{23}$) atoms or molecules. It has definite shape and size. The solids we come across in daily life can be classified into three groups :
 - Crystalline solids.
 - Amorphous solids.
 - Semi-crystalline solids.
- Crystalline solids.** Those solids in which the atoms or molecules are arranged in a regular and repeated geometrical pattern are called crystalline solids. Such solids are bounded by flat surfaces, are anisotropic, have sharp melting points and have long range order in their structure.

Examples. Rock salt, quartz, mica, calcite, diamond, etc.
- Amorphous solids.** These are the solids in which the atoms or molecules are not arranged according to certain definite geometrical order, *i.e.*, the atoms or molecules are arranged in a random order. Such solids are isotropic, do not have flat surfaces and their melting points are not sharp. They are super-cooled liquids.

Examples. Glass, rubber, cellulose, bitumen, bone and many plastics.
- Semi-crystalline solids.** These are the solids in which the crystalline phase is inter-dispersed in the amorphous phase, *i.e.*, in which crystalline and amorphous phases co-exist. Polyethylene and protein are such solids.
- Deforming force.** A force which changes the size and shape of body is called deforming force.
- Elasticity.** The property by virtue of which a body regains its original size and shape after the removal of deforming force is called elasticity.
- Perfectly elastic body.** If a body regains its original size and shape completely and immediately after the removal of deforming force, it is said to be perfectly elastic body. The nearest approach to a perfectly elastic body is quartz fibre.
- Plasticity.** The property by virtue of which a body does not regain its original size and shape even after the removal of the deforming force is called plasticity.
- Perfectly plastic body.** If a body does not show any tendency to regain its original size and shape even after the removal of deforming force, it is said to be perfectly plastic body. Putty and paraffin wax are nearly perfectly elastic bodies.
- Stress.** The restoring force set up per unit area of a deformed body is called stress.
$$\text{Stress} = \frac{\text{Restoring force}}{\text{Area}} = \frac{\text{Applied force}}{\text{Area}} = \frac{F}{A}$$

The SI unit of stress is Nm^{-2} and the CGS unit is dyne cm^{-2} . Its dimensional formula is $[\text{ML}^{-1}\text{T}^{-2}]$.
Stress is of two types :
 - Normal stress
 - Tangential stress.

13. **Strain.** The ratio of the change in any dimension produced in the body to the original dimension is called strain.

$$\text{Strain} = \frac{\text{Change in dimension}}{\text{Original dimension}}$$

As strain is the ratio of two like quantities, it has no units and dimensions. Strain is of *three* types :

- (i) **Longitudinal strain.** It is defined as the ratio of change in length to the original length.

$$\text{Longitudinal strain} = \frac{\text{Change in length}}{\text{Original length}} = \frac{\Delta l}{l}$$

- (ii) **Volumetric strain.** It is defined as the ratio of the change in volume to the original volume.

$$\text{Volumetric strain} = \frac{\text{Change in volume}}{\text{Original volume}} = \frac{\Delta V}{V}$$

- (iii) **Shear strain.** It is defined as the angle θ (in radian) through which a face originally perpendicular to the fixed face gets turned on applying tangential deforming force.

$$\text{Shear strain} = \theta = \tan \theta$$

$$= \frac{\text{Relative displacement between two } \parallel \text{ planes}}{\text{Distance between } \parallel \text{ planes}} \\ = \frac{\Delta l}{l}$$

14. **Elastic limit.** The maximum stress upto which stress is proportional to strain is called elastic limit.

15. **Hooke's law.** It states that within the elastic limit, stress is proportional to strain.

$$\text{Stress} \propto \text{Strain}$$

$$\text{or } \text{Stress} = E \times \text{strain}$$

The constant E is called modulus of elasticity of the material of the body.

16. **Modulus of elasticity or coefficient of elasticity.** It is defined as the ratio of stress to the corresponding strain, within the elastic limit.

It is of *three* types :

- (i) Young's modulus

- (ii) Bulk modulus

- (iii) Modulus of rigidity.

17. **Young's modulus of elasticity.** It is defined as the ratio of longitudinal stress to the longitudinal strain within the elastic limit.

It is given by

$$Y = \frac{\text{Longitudinal stress}}{\text{Longitudinal strain}} = \frac{F/A}{\Delta l/l} = \frac{F}{A} \cdot \frac{l}{\Delta l}$$

18. **Bulk modulus of elasticity.** It is defined as the ratio of normal stress to volumetric strain within the elastic limit.

It is given by

$$\kappa = \frac{\text{Normal stress}}{\text{Volumetric strain}} = - \frac{F/A}{\Delta V/V} \\ = - \frac{p}{\Delta V/V} = - \frac{pV}{\Delta V}$$

The negative sign shows that the volume decreases with the increase in stress.

19. **Modulus of rigidity.** It is defined as the ratio of tangential stress to shear strain within the elastic limit.

It is given by

$$\eta = \frac{\text{Tangential stress}}{\text{Shear strain}} = \frac{F/A}{\theta} = \frac{F/A}{\Delta l/l} = \frac{F}{A} \cdot \frac{l}{\Delta l}$$

20. **Units of moduli of elasticity.** As strain is a pure ratio, the unit of elasticity is same as that of stress. So SI unit of Y , κ or η is Nm^{-2} and the CGS unit is dyne cm^{-2} .

21. **Compressibility.** The reciprocal of the bulk modulus of a material is called its compressibility.

$$\text{Compressibility} = 1/\kappa$$

The SI unit of compressibility is $\text{N}^{-1} \text{m}^2$ or Pa^{-1} .

22. **Yield point.** The stress beyond which a solid flows is called yield point. For example, a paste of flour and water flows under its own weight.

23. **Breaking stress.** The stress corresponding to which a wire breaks is called breaking stress.

$$\text{Breaking force} = \text{Breaking stress} \times \text{area of cross-section of the wire.}$$

24. **Plastic region.** The region of stress-strain curve between the elastic limit and the breaking point is called plastic region.

25. **Ductile materials.** The materials which have large plastic range of extension are called ductile materials. Such materials can be drawn into thin wires.

Examples. Copper, silver, iron, aluminium, etc.

26. **Brittle materials.** The materials which have very small range of plastic extension are called brittle materials. Such materials break as soon as the stress is increased beyond the elastic limit.

Examples. Cast iron, glass, ceramics, etc.

27. **Malleable metals.** The metals which can be hammered or rolled into thin sheets are called malleable metals.

Examples. Gold, silver, lead, etc.

28. **Elastomers.** The materials which can be elastically stretched to large values of strain are called elastomers. They have large elastic region but do not obey Hooke's law.

Examples. Rubber and elastic tissue of aorta.

29. **Elastic after effect.** The delay in regaining the original state by a body on the removal of deforming force is called elastic after effect. This effect is minimum for quartz and phosphor bronze and maximum for glass fibre.
30. **Elastic fatigue.** It is defined as the loss in the strength of a material caused due to repeated alternating strains to which the material is subjected.
31. **Elastic hysteresis.** The fact that the stress-strain curve is not retraced on reversing the strain (for a material like rubber) is called elastic hysteresis.
32. **Poisson's ratio.** When the length of a loaded wire increases, its diameter decreases. The ratio of the lateral strain to the longitudinal strain is called Poisson's ratio. It is given by

$$\sigma = \frac{\text{Lateral strain}}{\text{Longitudinal strain}}$$

$$= - \frac{\Delta D / D}{\Delta l / l}$$

Poisson's ratio σ is a pure number. It has no units or dimensions.

33. **Elastic potential energy in a stretched wire.** The work done against the internal restoring forces in stretching a wire is stored as its elastic potential energy.

It is given by

$$U = \frac{1}{2} \text{Stretching force} \times \text{extension}$$

$$= \frac{1}{2} F \Delta l$$

$$= \frac{1}{2} \cdot \frac{F}{A} \cdot \frac{\Delta l}{l} \cdot A l$$

$$= \frac{1}{2} \text{Stress} \times \text{strain} \times \text{volume of the wire}$$

P.E. stored per unit volume of the wire or elastic energy density is

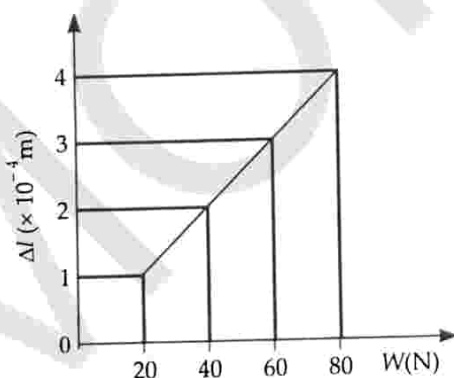
$$u = \frac{U}{V} = \frac{1}{2} \text{Stress} \times \text{strain}$$

$$= \frac{1}{2} \text{Young's modulus} \times \text{strain}^2.$$

IIT Entrance Exam

★ MULTIPLE CHOICE QUESTIONS WITH ONE CORRECT ANSWER

1. The adjacent graph shows the extension (Δl) of a wire of length 1 m suspended from the top of a roof at one end and with a load W connected to the other end.



If the cross-sectional area of the wire is 10^{-6} m^2 , calculate the Young's modulus of the material of the wire

- (a) $2 \times 10^{11} \text{ N/m}^2$
 (b) $2 \times 10^{11} \text{ N/m}^2$
 (c) $3 \times 10^{12} \text{ N/m}^2$
 (d) $2 \times 10^{13} \text{ N/m}^2$

[IIT 04]

2. The following four wires are made of the same material. Which of these will have the largest extension, when the same tension is applied?

- (a) length = 50 cm, diameter = 0.5 mm
 (b) length = 100 cm, diameter = 1 mm
 (c) length = 200 cm, diameter = 2 mm
 (d) length = 300 cm, diameter = 3 mm [IIT 81]

3. A wire of length L , and cross-sectional area A is made of a material of Young's modulus Y . If the wire is stretched by an amount x , the work done is

- (a) $YAx^2 / 2L$ (b) YAx^2 / L
 (c) $YAx / 2L$ (d) $YAx^2 L$ [IIT 87]

4. The pressure of a medium is changed from $1.01 \times 10^5 \text{ Pa}$ to $1.65 \times 10^5 \text{ Pa}$ and change in volume is 10% keeping temperature constant. The bulk modulus of the medium is

- (a) $204.8 \times 10^5 \text{ Pa}$ (b) $102.4 \times 10^5 \text{ Pa}$
 (c) $51.2 \times 10^5 \text{ Pa}$ (d) $1.55 \times 10^5 \text{ Pa}$ [IIT 05]

5. A given quantity of an ideal gas is at pressure P and absolute temperature T . The isothermal bulk modulus of the gas is

- (a) $2P/3$ (b) P
 (c) $3P/2$ (d) $2P$ [IIT 98]

Mechanical Properties of Fluids

GLIMPSES

- Fluid.** A fluid is a substance that can flow. The term fluid refers to both liquids and gases.
- Fluid statics.** The branch of physics that deals with the study of fluids at rest is called *fluid statics* or *hydrostatics*.
- Fluid dynamics.** The branch of physics that deals with the study of fluids in motion is called *fluid dynamics* or *hydrodynamics*.
- Thrust.** The total force exerted by a liquid on any surface in contact with it is called thrust. A liquid always exerts force perpendicular to the surface of the container at every point.
- Pressure.** The thrust exerted by a liquid per unit area of the surface in contact with it is known as pressure.

$$\text{Pressure} = \frac{\text{Thrust}}{\text{Area}} \quad \text{or} \quad P = \frac{F}{A}$$

Pressure is a scalar quantity.

- Units and dimensions of pressure.** The CGS unit of pressure is dyne cm^{-2} and its SI unit is Nm^{-2} which is also called pascal (Pa). The dimensional formula of pressure is $[\text{ML}^{-1}\text{T}^{-2}]$.
- Density.** The density of any material is defined as its mass per unit volume.

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}} \quad \text{or} \quad \rho = \frac{M}{V}$$

Density is a positive scalar quantity.

- Units and dimensions of density.** The SI unit of density is kg m^{-3} and the CGS unit is g cm^{-3} . The dimensional formula of density is $[\text{ML}^{-3}]$.
- Specific gravity.** The relative density or *specific gravity* of a substance is defined as the ratio of the density of the substance to the density of water at 4°C .

$$\text{Specific gravity} = \frac{\text{Density of substance}}{\text{Density of water at } 4^\circ\text{C}}$$

Specific gravity is a dimensionless positive scalar quantity.

- Pascal's law.** It states that a change in pressure applied to an enclosed incompressible fluid is transmitted undiminished to every point of the fluid and the walls of the containing vessel. Or, the pressure exerted at any point on an enclosed liquid is transmitted equally in all directions.
- Hydraulic lift.** It is an application of Pascal's law. It is used to lift heavy objects.

According to Pascal's law,

Pressure applied on smaller piston

= Pressure transmitted to larger piston

$$P = \frac{f}{a} = \frac{F}{A} \quad \text{or} \quad F = P \times A = \frac{f}{a} \times A$$

As $A > a$, so $F > f$.

Thus hydraulic lift acts as a force multiplier.

- Hydraulic brakes.** The hydraulic brakes used in automobiles are based on Pascal's law of transmission of pressure in a liquid.
- Pressure exerted by a liquid.** A liquid column of height h and density ρ exerts a pressure given by

$$P = h \rho g$$
- Effect of gravity on fluid pressure.** The pressure in a fluid varies with depth h according to the expression

$$P = P_a + h \rho g$$
 where ρ is the fluid density, assumed uniform.
- Hydrostatic paradox.** The pressure exerted by a liquid column depends only on the height of the liquid column and not on the shape of the containing vessel.

16. **Atmospheric pressure.** The pressure exerted by the atmosphere is called atmospheric pressure. At sea-level, we have

Atmospheric pressure

= Pressure exerted by 0.76 m of Hg

$$= h \rho g = 0.76 \times 13.6 \times 10^3 \times 9.8 = 1.013 \times 10^5 \text{ Nm}^{-2}$$

Other units used for atmospheric pressure are as follows :

$$1 \text{ atm} = 1.013 \times 10^6 \text{ dyne cm}^{-2}$$

$$= 1.013 \times 10^5 \text{ Nm}^{-2} \text{ (or Pa)}$$

$$1 \text{ bar} = 10^6 \text{ dyne cm}^{-2} = 10^5 \text{ Nm}^{-2}$$

$$1 \text{ millibar (m bar)} = 10^{-3} \text{ bar} = 10^3 \text{ dyne cm}^{-2} = 10^2 \text{ Nm}^{-2}$$

$$1 \text{ torr} = 1 \text{ mm Hg}$$

$$1 \text{ atm} = 101.3 \text{ kPa} = 1.013 \text{ bar} = 760 \text{ torr}$$

17. **Absolute pressure and gauge pressure.** The total or actual pressure P at a point is called absolute pressure. Gauge pressure is the difference between the actual pressure (absolute pressure) at a point and the atmospheric pressure. Thus

$$P_g = P - P_a \quad \text{or} \quad P = P_a + P_g$$

Absolute pressure = Atmospheric pressure
+ Gauge pressure.

18. **Buoyancy and centre of buoyancy.** The upward force acting on a body immersed in a fluid is called upthrust or buoyant force and the phenomenon is called buoyancy. The force of buoyancy acts through the centre of gravity of the displaced fluid which is called centre of buoyancy.

19. **Archimedes' principle.** It states that when a body is immersed partly or wholly in a fluid, it loses some weight. The loss in weight is equal to the weight of the fluid displaced.

Apparent weight of a body in a fluid

$$= \text{True weight} - \text{Weight of fluid displaced}$$

$$W_{app} = W - U = V \sigma g - V \rho g$$

$$= V \sigma g \left(1 - \frac{\rho}{\sigma} \right) = W \left(1 - \frac{\rho}{\sigma} \right)$$

where $W = V \sigma g$ is the weight of the body and σ its density.

20. **Law of floatation.** A body will float in a liquid if weight of the liquid displaced by the body is atleast equal to or greater than the weight of the body.

When a body just floats,

Weight of the body = Weight of liquid displaced

$$V \sigma g = V' \rho g \quad \text{or} \quad \frac{V'}{V} = \frac{\sigma}{\rho}$$

or $\frac{\text{Volume of the immersed part}}{\text{Total volume of the body}}$

$$= \frac{\text{Density of the body}}{\text{Density of liquid}}$$

21. **Viscosity.** It is the property of a fluid due to which an opposing force comes into play whenever there is relative motion between its different layers.
22. **Newton's formula for viscous force.** The viscous drag between two parallel layers each of area A and having velocity gradient dv/dx is given by

$$F = -\eta A \frac{dv}{dx}$$

where η is the coefficient of viscosity of the liquid.

23. **Coefficient of viscosity.** It may be defined as the tangential viscous force required to maintain a unit velocity gradient between two liquid layers each of unit area. Its dimensional formula is $[ML^{-1} T^{-1}]$.

24. **Units of η .** The CGS unit of η is *poise*. The coefficient of viscosity of a liquid is 1 poise if a tangential force of 1 dyne cm^{-2} of the surface is required to maintain a relative velocity of 1 cm s^{-1} between two layers of the liquid 1 cm apart.

$$1 \text{ poise} = 1 \text{ dyne s cm}^{-2} = 1 \text{ g cm}^{-1} \text{ s}^{-1}.$$

The SI unit of η is *decapoise*. The coefficient of viscosity of a liquid is 1 decapoise if a tangential force of 1 Nm^{-2} of the surface is required to maintain a relative velocity of 1 ms^{-1} between two layers of the liquid 1 m apart.

$$1 \text{ poiseuille} = 1 \text{ decapoise} = 1 \text{ Nsm}^{-2}$$

$$= 1 \text{ Pa s} = 10 \text{ poise}.$$

25. **Poiseuille's formula.** The volume of a liquid flowing per second through a horizontal capillary tube of length l , radius r under a pressure difference p across its two ends is given by

$$Q = \frac{V}{t} = \frac{\pi p r^4}{8 \eta l}.$$

26. **Stokes' law.** It states that the backward dragging force of viscosity acting on a spherical body of radius r moving with velocity v through a fluid of viscosity η is $F = 6\pi\eta rv$.

27. **Terminal velocity.** It is the maximum constant velocity attained by a spherical body while falling through a viscous medium. The terminal velocity of a spherical body of density ρ and radius r moving through a fluid of density ρ' and viscosity η is given by $v = \frac{2}{9} \frac{r^2}{\eta} (\rho - \rho') g$

28. **Streamline flow and turbulent flow.** It is the flow of liquid in which each particle of the liquid passing through a point travels along the same

path and with the same velocity as the preceding particle passing through the same point.

A liquid possesses streamline motion only when its velocity is less than a certain limiting value, called *critical velocity*. When the velocity of the liquid becomes greater than the critical velocity, the particles follow zig-zag path, such a disordered or irregular motion is called turbulent flow.

29. **Tube of flow.** A tube of flow is a bundle of streamlines having the same velocity of fluid element over any cross-section perpendicular to the direction of flow.

30. **Laminar flow.** The steady flow in which liquid moves in the form of layers is called laminar flow. The velocity of the layer varies from maximum at axis to zero for the layer at the wall of the tube.

31. **Critical velocity.** The critical velocity of a liquid is that limiting value of its velocity of flow upto which the flow is streamlined and above which the flow becomes turbulent. It is given by

$$v_c = \frac{k\eta}{\rho D}$$

32. **Reynold's number.** It is a dimensionless number which determines the nature of the flow of the liquid. For a liquid of viscosity η , density ρ and flowing through a pipe of diameter D , Reynold's number is given by

$$R_e = \frac{\rho v D}{\eta}$$

If $R_e < 2000$, the flow is laminar.

If $R_e > 3000$, the flow is turbulent.

If $2000 < R_e < 3000$, the flow is unstable. It may change from laminar to turbulent and vice-versa.

$$\text{Reynold's number} = \frac{\text{Inertial force per unit area}}{\text{Viscous force per unit area}}$$

33. **Ideal fluid.** An ideal fluid is one which is non-viscous, incompressible, and its flow is steady and irrotational.

34. **Rate of flow.** The volume of a liquid flowing per second through a pipe of cross-section a with velocity v is given by

$$Q = \frac{V}{t} = av.$$

35. **Equation of continuity.** If there is no source or sink of the fluid along the length of the pipe, the mass of the fluid crossing any section of the pipe per second is always constant.

$$m = a_1 v_1 \rho_1 = a_2 v_2 \rho_2$$

It is called equation of continuity. For an incompressible liquid, $\rho_1 = \rho_2$, then

$$a_1 v_1 = a_2 v_2 \quad \text{or} \quad av = \text{constant}.$$

Thus during the streamlined flow of a non-viscous and incompressible fluid through a pipe of varying cross-section, the product of area of cross-section and the normal fluid velocity remains constant throughout the fluid flow.

36. **Bernoulli's principle.** It states that the sum of pressure energy, kinetic energy and potential energy per unit volume of an incompressible, non-viscous fluid in a streamlined, irrotational flow remains constant along a streamline. Thus

$$P + \frac{1}{2} \rho v^2 + \rho gh = \text{constant}$$

$$\text{or} \quad \frac{P}{\rho g} + h + \frac{1}{2} \frac{v^2}{g} = \text{constant}$$

The terms $\frac{P}{\rho g}$, h and $\frac{v^2}{2g}$ are called pressure head, gravitational head and velocity head respectively. For the horizontal flow of a liquid ($h = \text{constant}$), Bernoulli's equation takes the form

$$P + \frac{1}{2} \rho v^2 = \text{constant}$$

$$\text{or} \quad P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2.$$

It indicates that velocity increases where pressure decreases and vice-versa.

37. **Torricelli's Law.** It states that the velocity of efflux i.e. the velocity with which the liquid flows out of an orifice (a narrow hole) is equal to that which a freely falling body would acquire in falling through a vertical distance equal to the depth of orifice below the free surface of liquid. Hence the velocity of efflux of a liquid through an orifice at depth h from the liquid surface will be

$$v = \sqrt{2gh}.$$

38. **Venturimeter.** It is a device used to measure the rate of flow of a liquid through a pipe. It is an application of Bernoulli's principle. It is also called flow meter or venturi tube. Volume of the liquid flowing out per second is given by

$$Q = a_1 a_2 \sqrt{\frac{2h \rho_m g}{\rho (a_1^2 - a_2^2)}}.$$

39. **Magnus effect.** The difference in lateral pressure, which causes a spinning ball to take a curved path which is convex towards the greater pressure side, is called magnus effect.

40. **Aerofoil.** It is a solid object shaped to provide an upward vertical force as it moves horizontally through air.

41. **Cohesive and adhesive forces.** The force of attraction between the molecules of the same substance is called cohesive force while the force of attraction between the molecules of two different substances is called adhesive force.

42. **Molecular range.** It is the maximum distance upto which a molecule can exert some measurable attraction on other molecules. The order of molecular range is 10^{-9} m in solids and liquids.
43. **Sphere of influence.** It is a sphere drawn with molecule as centre and molecular range as radius.
44. **Surface film.** A thin film of liquid near its surface and having thickness equal to the molecular range for that liquid is called surface film. The molecules present in the surface film possess additional potential energy.
45. **Surface Tension.** It is the property of a liquid by virtue of which, it behaves like an elastic stretched membrane with a tendency to contract so as to occupy a minimum surface area. It is measured as the force per unit length on an imaginary line drawn on the surface of liquid.

$$\text{Surface tension} = \frac{\text{Force}}{\text{Length}} \text{ or } \sigma = \frac{F}{l}$$

Its SI unit is Nm^{-1} and CGS unit is dyne cm^{-1} .

46. **Surface energy.** The additional potential energy per unit area of the surface film as compared to the molecules in the interior is called the surface energy.

Surface energy

$$= \frac{\text{Work done in increasing the surface area}}{\text{Increase in surface area}}$$

Surface energy of a liquid is numerically equal to surface tension of the liquid.

47. **Excess pressure inside a drop and bubble.** There is excess of pressure on the concave side of a curved surface.

$$\text{Excess pressure inside a liquid drop} = \frac{2\sigma}{R}$$

(One free surface)

$$\text{Excess pressure inside a liquid bubble} = \frac{4\sigma}{R}$$

(Two free surfaces)

$$\text{Excess pressure inside an air bubble} = \frac{2\sigma}{R}$$

(One free surface)

where R is radius of the liquid drop, liquid bubble or the air bubble.

48. **Angle of contact.** The angle, which the tangent to the free surface of liquid at the point of contact makes with the wall of the containing vessel inside the liquid, is called angle of contact. For the liquids having concave meniscus, the angle of contact is acute and for the liquids having convex meniscus, the angle of contact is obtuse. The liquids, for which the angle of contact is acute, show a rise in the capillary tube; while those for which the angle of contact is obtuse, show a fall.
49. **Capillarity.** A tube of very fine bore is called a capillary tube. The phenomenon of rise or fall of a liquid in capillary tube is known as capillarity.
50. **Ascent formula.** When a capillary tube of radius r is dipped in a liquid of density ρ and surface tension σ , the liquid rises or falls through a distance,

$$h = \frac{2\sigma \cos \theta}{r \rho g}$$

51. **Rise of liquid in a tube of insufficient height.** The radius r of the capillary tube and radius of curvature R of the liquid meniscus are related by $r = R \cos \theta$. Therefore

$$h = \frac{2\sigma \cos \theta}{R \cos \theta \cdot \rho g} = \frac{2\sigma}{R \rho g}$$

As, σ , ρ and g are constants, so

$$hR = \frac{2\sigma}{\rho g} = \text{a constant}$$

If $h' < h$, then the radius of curvature R increases to R' such that $hR = h' R'$. The liquid rises and spreads out to a new radius $R' = hR / h'$. But the liquid does not overflow.

IIT Entrance Exam

MULTIPLE CHOICE QUESTIONS WITH ONE CORRECT ANSWER

1. A closed compartment containing gas is moving with some acceleration in horizontal direction. Neglect effect of gravity. Then the pressure in the compartment is

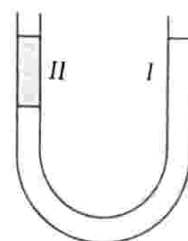
- same everywhere
- lower in the front side
- lower in the rear side
- lower in the upper side

[IIT 99]

2. A U-tube of uniform cross-section is partially filled with a liquid I. Another liquid II which does not mix with liquid I is poured into one side. It is found that the liquid levels of the two sides of the tube are the same, while the level of liquid I has risen by 2 cm. If the specific gravity of liquid I is 1.1, the specific gravity of liquid II must be

- 1.12
- 1.1
- 1.05
- 1.0

[IIT 83]



Competition Section

Thermal Properties of Matter

GLIMPSES

1. **Heat.** Heat is a form of energy which produces in us the sensation of hotness or coldness. According to *dynamic theory*, heat may be regarded as the energy of molecular motion which is equal to the sum total of the kinetic energy possessed by the molecules by virtue of their translational, vibrational and rotational motions.

2. **Units of heat.** Calorie (cal) is the C.G.S. unit of heat. One calorie is defined as the heat energy required to raise the temperature of one gram of water from 14.5°C to 15.5°C. Like all other forms of energy, the S.I. unit of heat is joule.

$$1 \text{ calorie} = 4.186 \text{ joule.}$$

3. **Joule's mechanical equivalent of heat.** Whenever a given amount of work (W) is converted into heat, always the same amount of heat Q is produced.

$$\text{Thus } W \propto Q$$

$$\text{or } W = JQ$$

$$\text{or } J = \frac{W}{Q}$$

The proportionality constant J is called Joule's mechanical equivalent of heat. It may be defined as the amount of work that must be done to produce a unit quantity of heat.

$$J = 4.2 \times 10^7 \text{ erg cal}^{-1} = 4.2 \text{ J cal}^{-1}.$$

4. **Temperature.** It is the degree of hotness of a body. The temperature of a body gives a measure of the average kinetic energy of its molecules.

5. **Thermometer.** It is a device used to measure the temperature of a body. It makes use of some measurable property (called thermometric property) of a substance which changes linearly with temperature.

6. **Different temperature scales.**

Temperature scale	Lower fixed point (Melting point of ice)	Upper fixed point (Boiling point of water)
1. Celsius	0°C	100°C
2. Fahrenheit	32°F	212°F
3. Reaumer	0°R	80°R
4. Kelvin	273.15 K	373.15 K

7. **Relations between different temperature scales.** If T_C , T_F , T_R and T are the temperatures of a body on Celsius, Fahrenheit, Reaumer and Kelvin scales respectively, then

$$(i) \frac{T_C - 0}{100 - 0} = \frac{T_F - 32}{212 - 32} = \frac{T_R - 0}{80 - 0} = \frac{T - 273.15}{373.15 - 273.15}$$

$$\text{or } \frac{T_C}{5} = \frac{T_F - 32}{9} = \frac{T_R}{4} = \frac{T - 273.15}{5}$$

$$(ii) T_C = \frac{5}{9} (T_F - 32), \quad T_F = \frac{9}{5} T_C + 32$$

$$(iii) T = T_C + 273.15, \quad T_C = T - 273.15$$

$$(iv) T_F = \frac{9}{5} (T - 273.15) + 32 = \frac{9}{5} T - 459.67,$$

$$T = \frac{5}{9} T_F + 255.37.$$

8. **Absolute scale of temperature.** The lowest possible temperature of -273.15°C at which a gas is supposed to have zero volume (and zero pressure) and at which entire molecular motion stops is called absolute zero of temperature. The temperature scale which starts with -273.15°C as its zero is called Kelvin scale or absolute scale. The size of degree on Kelvin scale is same as that on Celsius scale.

$$T (\text{K}) = t^\circ (\text{C}) + 273.15.$$

9. **Triple point of water.** The triple point of water is the state at which the three phases of water namely ice, liquid water and water vapour are equally stable and co-exist in equilibrium. It is unique because it occurs at a specific temperature of 273.16 K and a specific pressure of 0.46 cm of Hg column.

10. **Constant volume air thermometer.** It is used to measure the pressure of a definite mass of air at different temperatures, the volume of air remaining constant. It is based on the pressure law,

$$\frac{P}{T} = \frac{P_0}{T_0} \quad \text{or} \quad T = T_0 \times \frac{P}{P_0}$$

In terms of triple point of water, $T = T_{tr} \times \frac{P}{P_{tr}}$.

11. **Ideal gas temperature scale.** The ideal gas temperature on the Kelvin scale is defined by the equation

$$T = \lim_{P_{tr} \rightarrow 0} 273.16 \left(\frac{P}{P_{tr}} \right)$$

It is independent of the nature of the gas.

12. **Platinum resistance thermometer.** It is based on the fact that resistance of a platinum wire varies with temperature. If R_0 and R are the resistances at 0°C and $t^\circ\text{C}$ respectively, then

$$R = R_0 (1 + \alpha t)$$

Here α is the temperature coefficient of resistance of platinum and is defined as the increase in resistance per unit resistance at 0°C for 1°C rise in temperature.

$$\alpha = \frac{R - R_0}{R_0 \times t}$$

Unit of α is $^\circ\text{C}^{-1}$.

If R_0 and R_{100} are the resistances of a platinum wire at ice point and steam point respectively, then the temperature t_R of a body for which the corresponding resistance is R_t is given by

$$t_R = \frac{R_t - R_0}{R_{100} - R_0} \times 100^\circ\text{C}.$$

13. **Thermoelectric thermometer.** It uses thermoelectric e.m.f. ($\mathcal{E}$) as thermometric property. For the linear part of the thermo e.m.f., the unknown temperature is given by

$$t_\mathcal{E} = \frac{\mathcal{E}_t - \mathcal{E}_0}{\mathcal{E}_{100} - \mathcal{E}_0} \times 100 \text{ degrees}$$

14. **Linear expansion.** When a solid rod of initial length l is heated through a temperature ΔT , its final (increased) length is given by

$$l' = l (1 + \alpha \Delta T)$$

where α is coefficient of linear expansion. It is given by

$$\alpha = \frac{l' - l}{l \times \Delta T}$$

The coefficient of linear expansion of the material of a solid rod is defined as the increase in length per unit length per degree rise in temperature.

15. **Superficial expansion.** When a solid sheet of initial surface area S is heated through a temperature ΔT , its final (increased) surface area is given by

$$S' = S (1 + \beta \Delta T)$$

where β is coefficient of superficial expansion. It is given by

$$\beta = \frac{S' - S}{S \times \Delta T}$$

It is defined as the increase in surface area per unit surface area per degree rise in temperature.

16. **Cubical expansion.** When a solid of initial volume V is heated through a temperature ΔT , its final (increased) volume is given by $V' = V (1 + \gamma \Delta T)$ where γ is coefficient of cubical expansion. It is given by

$$\gamma = \frac{V' - V}{V \times \Delta T}$$

It is defined as the increase in volume per unit volume per degree rise in temperature.

The coefficient of cubical expansion of an ideal gas is equal to the reciprocal of its absolute temperature.

$$\gamma = \frac{1}{T}$$

17. **Relation between α , β and γ .** The three coefficients of thermal expansion are related as

$$\frac{\alpha}{1} = \frac{\beta}{2} = \frac{\gamma}{3} \quad \text{or} \quad \beta = 2\alpha \quad \text{and} \quad \gamma = 3\alpha.$$

The units of α , β and γ are same viz. $^\circ\text{C}^{-1}$ or K^{-1} .

18. **Coefficient of apparent expansion of a liquid.** It is defined as the apparent increase in volume per unit original volume per degree rise in temperature.

$$\gamma_a = \frac{\text{Apparent increase in volume}}{\text{Original volume} \times \text{Rise in temperature}}$$

19. **Coefficient of real expansion of a liquid.** It is defined as the real increase in volume per unit original volume per degree rise in temperature.

$$\gamma_r = \frac{\text{Real increase in volume}}{\text{Original volume} \times \text{Rise in temperature}}$$

If γ_g is the coefficient of cubical expansion of glass (material of the container), then

$$\gamma_r = \gamma_a + \gamma_g$$

20. **Variation of density with temperature.** The density of a solid or liquid decreases with the increase of temperature.

$$\rho' = \rho (1 - \gamma \Delta T)$$

21. **Specific heat.** It may be defined as the amount of heat required to raise the temperature of unit mass of a substance through one degree. If Q heat is needed to raise the temperature of m mass of a substance through ΔT , then specific heat is

$$c = \frac{Q}{m\Delta T}$$

The cgs unit of specific heat is $\text{cal g}^{-1}^{\circ}\text{C}^{-1}$ and the SI unit is $\text{J kg}^{-1} \text{K}^{-1}$. Clearly,

Heat gained or lost, $Q = mc \Delta T$.

22. **Molar specific heat.** It is defined as the amount of heat required to raise the temperature of 1 mole of the substance through one degree. If Q heat is needed to raise the temperature of n moles of a substance through ΔT , then molar specific heat is

$$C = \frac{Q}{n\Delta T}$$

The cgs unit of molar specific heat is $\text{cal mol}^{-1}^{\circ}\text{C}^{-1}$ and the SI unit is $\text{J mol}^{-1} \text{K}^{-1}$.

23. **Heat capacity or thermal capacity.** It is defined as the amount of heat required to raise the temperature of a body through one degree.

Heat capacity = Mass $\times$ Specific heat = mc

The cgs unit of heat capacity is $\text{cal}^{\circ}\text{C}^{-1}$ and the S.I. unit is JK^{-1} .

24. **Water equivalent.** The water equivalent of a body is defined as the mass of water which requires the same amount of heat as is required by the given body for the same rise of temperature.

$w = \text{Mass} \times \text{specific heat} = mc$

The cgs unit of water equivalent is g and the SI unit is kg.

25. **Latent heat.** The amount of heat required to change the state of unit mass of a substance at a constant temperature is called its latent heat. It is denoted by L .

26. **Latent heat of fusion.** The amount of heat required to change the state of unit mass of a substance from solid to liquid at its melting point is called latent heat of fusion.

27. **Latent heat of vaporisation.** The amount of heat required to change the state of unit mass of a substance from liquid to vapour at its boiling point is called latent heat of vaporisation.

28. **Principle of calorimetry.** When two bodies at different temperatures are placed in contact with each other, the heat lost by the hot body is equal to the heat gained by the cold body. This is the principle of calorimetry or the principle of mixtures.

Heat gained = Heat lost

29. **Transfer of heat.** The three modes of transfer of heat are conduction, convection and radiation.

30. **Conduction.** It is a process in which heat is transmitted from one part of a body to another at a lower temperature through molecular collisions, without any actual flow of matter.

31. **Steady state.** The state of the rod when temperature of every cross-section of the rod becomes constant and there is no further absorption of heat in any part is called steady state.

32. **Factors on which conduction of heat depends.** When two opposite faces of a slab each of area of cross-section A and separated by a distance x are maintained at temperatures T_1 and T_2 ($T_1 > T_2$), then amount of heat that flows in time t ,

$$Q = \frac{KA(T_1 - T_2)t}{x}$$

where K is coefficient of thermal conductivity of the material of the slab between its two faces. The rate of flow of heat through the slab is

$$\frac{dQ}{dt} = -KA \frac{dT}{dx}$$

Here dT/dx is the rate of fall of temperature with distance and is called temperature gradient.

33. **Coefficient of thermal conductivity.** It may be defined as the quantity of heat energy that flows in unit time between the opposite faces of a cube of unit side, the faces being kept at one degree difference of temperature. Its cgs unit is $\text{cal s}^{-1}\text{cm}^{-1}^{\circ}\text{C}^{-1}$ and SI unit is $\text{Js}^{-1}\text{m}^{-1}\text{K}^{-1}$ or $\text{Wm}^{-1}\text{K}^{-1}$.

Dimensional formula of K is $[\text{MLT}^{-3}\text{K}^{-1}]$.

Thermal conductivities of metals are much greater than those for metals. Gases are poor thermal conductors.

34. **Heat current and resistance.** The flow of heat per unit time in conduction is called *heat current*. The ratio of the temperature difference between the ends of a conductor to the heat current through it is called *thermal resistance*.

Heat current, $H = \frac{Q}{t} = KA \frac{\Delta T}{\Delta x}$.

Thermal resistance, $R_H = \frac{\Delta T}{H} = \frac{\Delta x}{KA}$.

SI unit of $R_H = \text{KW}^{-1}$.

Dimensions of $R_H = [\text{M}^{-1}\text{L}^{-2}\text{T}^{-3}\text{K}]$.

35. **Convection.** It is the process by which heat is transmitted through a substance from one point to another due to the bodily motion of the heated particles of the substance. Fluids are mainly heated through convection. *Natural convection* arises due to unequal heating and gravity, when more heated and less dense parts rise and are replaced by the cooler parts of the fluid. In *forced convection*, a material is forced to move by an agency like a pump or a blower.

36. **Radiation.** It is the process by which heat is transmitted from one place to another without heating the intervening medium.

37. **Prevost's theory of heat exchanges.** All bodies emit radiations irrespective of their temperatures. They emit radiations to the surroundings and receive radiations from the surroundings. In the equilibrium state, the exchange of energy between a body and its surroundings occurs in equal amounts.

38. **Thermal radiation.** The electromagnetic radiation emitted by a body by virtue of its temperature is called thermal radiation or radiant energy. Thermal radiations are electromagnetic waves of long wavelength ranging from $1\mu\text{m}$ to $100\mu\text{m}$.

39. **Absorptive power or absorptivity (a_λ).** The absorptive power of a body for a given wavelength λ is defined as the ratio of amount of heat energy absorbed in a certain time to the total heat energy incident on it in the same time within a unit wavelength range around the wavelength λ . It is a dimensionless quantity.

40. **Emissive power (e_λ).** The emissive power of a body at a given temperature and for a given wavelength λ is defined as the amount of radiant energy emitted per unit time per unit surface area of the body within a unit wavelength range around the wavelength λ . Its SI unit is $\text{J s}^{-1} \text{m}^{-2}$ or W m^{-2} .

41. **Emissivity (ϵ).** It is the ratio of the emissive power of a body (e) to the emissive power (E) of a black body at the same temperature. It is given by

$$\epsilon = \frac{e}{E}$$

It is a dimensionless quantity having value between 0 and 1. The emissivity of a black body is 1.

42. **Black body.** A black body is one which neither reflects nor transmits but absorbs whole of the heat radiation incident on it. The absorptive power of a black body is unity.

43. **Kirchhoff's law.** It states that at any given temperature, the ratio of the emissive power (e_λ) to the absorptive power (a_λ) corresponding to certain wavelength is constant for all bodies and this constant is equal to the emissive power of the perfect black body (E_λ) at the same temperature and corresponding to the same wavelength. That is

$$\frac{e_\lambda}{a_\lambda} = E_\lambda \text{ (constant)}$$

Hence a good absorber is a good emitter.

44. **Stefan Boltzmann Law.** It states that the total amount of energy radiated per second per unit area of a perfect black body is directly proportional to the fourth power of the absolute temperature of the body.

$$\text{Thus } E \propto T^4 \quad \text{or} \quad E = \sigma T^4$$

where σ is called *Stefan's constant*. Its value is

$$\sigma = 5.67 \times 10^{-5} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ K}^{-4} \quad (\text{in CGS system})$$

$$\text{or } \sigma = 5.67 \times 10^{-8} \text{ Js}^{-1} \text{ m}^{-2} \text{ K}^{-4} \\ = 5.67 \times 10^{-8} \text{ Wm}^{-2} \text{ K}^{-4} \quad (\text{in SI})$$

If a perfect black body at temperature T is placed in an enclosure at temperature T_0 , then net amount of energy radiated per second per unit area by the black body is

$$E = \sigma(T^4 - T_0^4)$$

If the body and the enclosure are not perfect black bodies and have relative emissivity ϵ , then

$$E = \epsilon\sigma(T^4 - T_0^4)$$

45. **Newton's law of cooling.** For small temperature difference between a body and its surroundings, the rate of cooling of the body is directly proportional to the temperature difference and the surface area exposed. This is known as Newton's law of cooling.

$$\frac{dT}{dt} = -kA(T - T_0).$$

46. **Wien's displacement law.** It states that the wavelength (λ_m) corresponding to which the energy emitted by a perfect black body is maximum is inversely proportional to the absolute temperature (T) of the black body i.e.,

$$\lambda_m \propto \frac{1}{T} \quad \text{or} \quad \lambda_m = \frac{b}{T}$$

where b is a constant of proportionality and is called *Wien's constant*. Its value is

$$b = 2.9 \times 10^{-3} \text{ mK}$$

47. **Solar constant.** It is defined as the amount of solar radiant energy that a unit area of a perfect black body on the earth would receive per second in the absence of the atmosphere, with its surface perpendicular to the direction of the sun rays. Its value is 1340 W m^{-2} .

48. **Surface temperature of the sun.** If S_0 is solar constant, R_s the solar radius and R_0 is the mean distance of the earth from the sun, then surface temperature of the sun will be

$$T = \left[\frac{R_0^2 S_0}{R_s^2 \sigma} \right]^{1/4}$$

Thermodynamics

GLIMPSES

1. **Thermodynamics.** It is the branch of science that deals with the concepts of heat and temperature and the inter-conversion of heat and other forms of energy.
2. **Thermodynamic system.** An assembly of a very large number of particles having a certain value of pressure, volume and temperature is called a thermodynamic system.
3. **Surroundings.** Everything outside the system which can have a direct effect on the system is called its surroundings.
4. **Thermodynamic variables.** The quantities like pressure (P), volume (V) and temperature (T) which help us to study the behaviour of a thermodynamic system are called thermodynamic variables.
5. **Equation of state.** The mathematical relation between the pressure, volume and temperature of a thermodynamic system is called its equation of state. For example, the equation of state of n moles of an ideal gas can be written as

$$PV = nRT$$

6. **Thermal equilibrium.** Two systems are in thermal equilibrium with each other if they have the same temperature.
7. **Thermodynamic equilibrium.** A system is said to be in the state of thermodynamic equilibrium if the macroscopic variables describing the thermodynamic state of the system do not change with time. A system in a state of thermodynamic equilibrium possesses mechanical, thermal and chemical equilibria simultaneously.
8. **State variables.** The macroscopic quantities which are used to describe the equilibrium states of a thermodynamic system are called state variables. The value of a state variable depends only on the particular state, not on the path used to attain that

state. Pressure (P), volume (V), temperature (T) and mass (m) are state variables. Heat (Q) and work (W) are not state variables.

9. **Zeroth law of thermodynamics.** If two systems A and B are in thermal equilibrium with a third system C , then A and B are in thermal equilibrium with each other. According to this law, temperature is a physical quantity which has the same value for all systems which are in thermal equilibrium with each other.
10. **Internal energy.** The internal energy of a system is the sum of molecular kinetic and potential energies in the frame of reference relative to which the centre of mass of the system is at rest. It does not include the over-all kinetic energy of the system. It is a state variable denoted by U .
11. **Quasi-static process.** A quasi-static process is an infinitely slow process such that system remains in thermal and mechanical equilibrium with the surroundings throughout. In such a process, the pressure and temperature of the surroundings can differ from those of the system only infinitesimally.
12. **Isothermal process.** A process in which temperature remains constant is called an isothermal process. For such a process,
$$PV = \text{constant} \quad \text{or} \quad P_1 V_1 = P_2 V_2$$
13. **Adiabatic process.** A process in which thermally insulated system neither loses nor gains heat from the surroundings is called adiabatic process.

Equations of state for adiabatic processes are :

$$\begin{aligned} \text{(i)} \quad PV^\gamma &= \text{constant} & \text{or} & \quad P_1 V_1^\gamma = P_2 V_2^\gamma \\ \text{(ii)} \quad TV^{\gamma-1} &= \text{constant} & \text{or} & \quad T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1} \\ \text{(iii)} \quad \frac{P^{\gamma-1}}{T^\gamma} &= \text{constant} & \text{or} & \quad \frac{P_1^{\gamma-1}}{T_1^\gamma} = \frac{P_2^{\gamma-1}}{T_2^\gamma} \end{aligned}$$

where $\gamma = C_p / C_v$.

14. **Isobaric process.** A process in which volume remains constant is called isobaric process. For such a process

$$\frac{V}{T} = \text{constant} \quad \text{or} \quad \frac{V_1}{T_1} = \frac{V_2}{T_2}$$

15. **Isochoric process.** A process in which volume remains constant is called isochoric process. For such a process,

$$\frac{P}{T} = \text{constant} \quad \text{or} \quad \frac{P_1}{T_1} = \frac{P_2}{T_2}$$

16. **Indicator diagram.** A graphical representation of the state of a system with the help of two thermodynamical variables is called indicator diagram of the system. The graph between pressure P and volume V is called P - V diagram.

17. **Work done during the expansion of a gas.** When the volume of a gas changes from V_1 to V_2 , the work done is

$$W = \int_{V_1}^{V_2} P dV$$

= Area enclosed between the P - V curve and the volume axis.

18. **Work done during a cyclic process.** From the P - V diagram,

Work done per cycle = Area of the loop representing the cycle

- (i) If the loop is traced *clockwise*, the work done is *positive* and work is done by the system.
- (ii) If the loop is traced *anticlockwise*, the work done is *negative* and work is done on the system.

19. **First law of thermodynamics.** It states that if heat dQ is given to a system, a part of it is used in increasing the internal energy by an amount dU and the remaining energy is used in doing the external work dW . It is just a restatement of the law of conservation of energy. Thus

$$dQ = dU + dW \quad \text{or} \quad dQ = dU + PdV$$

Sign conventions used.

- (i) Heat absorbed by a system is positive and heat given out by a system is negative.
- (ii) Increase in internal energy of a system is positive and decrease in internal energy of a system is negative.
- (iii) Work done by a system is positive and work done on a system is negative.

20. **Work done in an isothermal process.** Work done when 1 mole of a gas expands isothermally,

$$W_{\text{iso}} = 2.303 RT \log \frac{V_2}{V_1} = 2.303 RT \log \frac{P_1}{P_2}$$

21. **Work done in an adiabatic process.** Work done when 1 mole of a gas expands adiabatically and its temperature falls from T_1 to T_2 ,

$$W_{\text{adia}} = \frac{R}{\gamma - 1} [T_1 - T_2] = \frac{1}{\gamma - 1} [P_1 V_1 - P_2 V_2]$$

22. **Dulong and Petit's law.** Near the room temperature, the molar specific heat of most of the solids at constant volume is equal to $3R$ or $6 \text{ cal mol}^{-1} \text{ K}^{-1}$ or $25 \text{ J mol}^{-1} \text{ K}^{-1}$. This statement is known as Dulong and Petit's law.

23. **Molar specific heat of a gas at constant volume (C_v).** It is defined as the amount of heat required to raise the temperature of 1 mole of the gas through 1°C at constant volume.

If c_v is specific heat of gas for 1 g at constant volume and M is its molecular weight, then molar specific heat at constant volume,

$$C_v = M c_v$$

24. **Molar specific heat of a gas at constant pressure (C_p).** It is defined as the amount of heat required to raise the temperature of 1 mole of the gas through 1°C at constant pressure. Thus,

$$C_p = M c_p$$

25. **Relation between two specific heats of a gas.** Specific heat of a gas at constant pressure is greater than the specific heat at constant volume.

For one mole of a gas :

$$(i) C_p - C_v = R \quad (\text{when } C_p, C_v \text{ are in units of work})$$

$$(ii) C_p - C_v = \frac{R}{J} \quad (\text{when } C_p, C_v \text{ are in units of heat})$$

where R is universal gas constant for 1 mole of a gas. For 1 g of a gas :

$$(i) c_p - c_v = r \quad (\text{when } c_p, c_v \text{ are in units of work})$$

$$(ii) c_p - c_v = \frac{r}{J} \quad (\text{when } c_p, c_v \text{ are in units of heat})$$

where $r = \frac{R}{M}$ = gas constant for 1 g of a gas.

Clearly, heat lost or gained by n moles of a gas,

$$(i) Q = n C_p \Delta T \quad (\text{At constant pressure})$$

$$(ii) Q = n C_v \Delta T \quad (\text{At constant volume})$$

where n = number of moles of gas

$$= \frac{\text{Mass of gas}}{\text{Molecular mass}}$$

26. **Heat engine.** It is a device which converts continuously heat energy into mechanical energy in a cyclic process. It essentially consists of (i) a source of heat (ii) a sink of heat and (iii) a working substance.

27. **Efficiency of a heat engine.** It is the ratio of useful work done (W) by the engine per cycle to the heat energy (Q) absorbed from the source per cycle.

$$\eta = \frac{\text{Work output}}{\text{Heat input}} = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

where Q_2 is the heat rejected to the sink.

28. **Second law of thermodynamics.**

- (i) **Kelvin-Planck statement.** It is impossible to construct an engine, which will produce no effect other than extracting heat from a reservoir and performing an equivalent amount of work.
- (ii) **Clausius statement.** It is impossible for a self acting machine, unaided by an external agency, to transfer heat from a body to another at higher temperature.

29. **Reversible process.** A process which can be made to proceed in the reverse direction by variation in its conditions so that any change occurring in any part of the direct process is exactly reversed in the corresponding part of the reverse process is called a reversible process.

30. **Irreversible process.** A process which cannot be made to proceed in the reverse direction is called an irreversible process.

31. **Carnot Engine.** It is an ideal heat engine which is based on Carnot's reversible cycle. Its working consists of four steps viz. Isothermal expansion, adiabatic expansion, isothermal compression and

adiabatic compression. The efficiency of Carnot engine is given by

$$\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$$

where T_1 and T_2 are the temperatures of source and sink respectively.

32. **Carnot theorem.** It states that

- (i) no engine working between two given temperatures can have efficiency greater than that of the Carnot engine working between the same two temperatures and
- (ii) the efficiency of the Carnot engine is independent of the nature of the working substance.

33. **Refrigerator.** It is a heat engine working in the reverse direction. Here a working substance absorbs heat Q_2 from the sink at temperature T_2 . An external agency does work W on the working substance. A larger amount of heat Q_1 is rejected to source at a higher temperature T_1 .

$$Q_1 = Q_2 + W.$$

34. **Coefficient of performance.** It is defined as the ratio of the amount of heat (Q_2) removed per cycle from the contents of the refrigerator to the work done (W) by the external agency to remove it.

$$\beta = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}.$$

IIT Entrance Exam

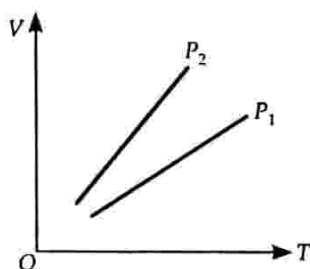
✓ MULTIPLE CHOICE QUESTIONS WITH ONE CORRECT ANSWER

1. In a given process of an ideal gas, $dW = 0$ and $dQ < 0$. Then for the gas

- (a) the temperature will decrease
(b) the volume will increase
(c) the pressure will remain constant
(d) the temperature will increase.

[IIT 01]

2. The volume (V) versus temperature (T) graphs for a certain amount of a perfect gas at two pressures P_1 and P_2 are shown in the figure. It follows from the graphs that

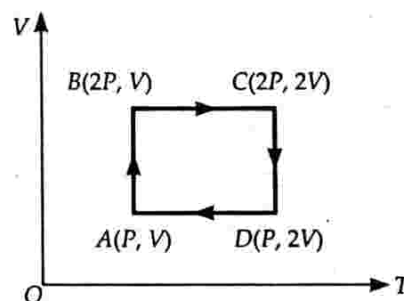


- (a) $P_1 > P_2$
(b) $P_1 < P_2$
(c) $P_1 = P_2$

- (d) information is insufficient to draw any conclusion.

[IIT 88]

3. An ideal monoatomic gas is taken round the cycle ABCDA as shown in the P - V diagram.



The work done during the cycle is

- (a) PV (b) $2PV$
(c) $PV/2$ (d) zero.

[IIT 83]

4. An ideal gas is taken through the cycle $A \rightarrow B \rightarrow C \rightarrow A$ as shown in the figure. If the net heat

Competition Section

Kinetic Theory of Gases

GLIMPSES

1. **Boyle's law.** It states that at constant temperature, the volume of a given mass of a gas is inversely proportional to its pressure.

$$V \propto 1/P \text{ or } PV = \text{constant}$$

$$\text{or } P_1 V_1 = P_2 V_2 \quad (\text{For constant } m \text{ and } T)$$

2. **Charles' law.** It states that if the pressure remains constant, then the volume of a given mass of a gas increases or decreases by $1/273.15$ of its volume at 0°C for each 1°C rise or fall of temperature. Mathematically,

$$V_t = V_0 \left(1 + \frac{t}{273.15} \right) = V_0 \left(\frac{273.15 + t}{273.15} \right)$$

If we put $273.15 + t = T$ and $273.15 = T_0$, then

$$\frac{V_t}{T} = \frac{V_0}{T_0} \text{ or } \frac{V}{T} = \text{constant}$$

So Charles' law can also be stated as follows :

At constant pressure, the volume of a given mass of a gas is directly proportional to its absolute temperature.

3. **Gay Lussac's law.** It states that if the volume remains constant, then the pressure of the given mass of a gas increases or decreases by $1/273.15$ of its pressure at 0°C for each 1°C rise or fall of temperature. Mathematically,

$$P_t = P_0 \left(1 + \frac{t}{273.15} \right) = P_0 \left(\frac{273.15 + t}{273.15} \right)$$

$$\text{or } \frac{P_t}{P_0} = \frac{T}{T_0}$$

$$\text{or } \frac{P}{T} = \text{constant}$$

So Gay Lussac's law can also be stated as follows :

At constant volume, the pressure of a given mass of a gas is directly proportional to its absolute temperature.

4. **Ideal gas equation.** For n moles of a gas,

$$PV = nRT \quad \text{or} \quad \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

For 1 mole of a gas,

$$PV = RT$$

5. **Universal gas constant.** It signifies the work done by the gas per mole per kelvin.

$$R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$$

6. **Boltzmann's constant.** It is the gas constant per molecule of a gas. If N is Avogadro's number, then

$$k_B = \frac{R}{N} = 1.38 \times 10^{-23} \text{ J molecule}^{-1} \text{ K}^{-1}$$

7. **Ideal or perfect gas.** A gas which obeys gas laws strictly is an ideal or perfect gas. The molecules of such a gas are of point size and there is no force of attraction between them.

The actual real gases obey the ideal gas equation only approximately at low pressures and high temperatures.

8. **Assumptions of kinetic theory of gases :**

(i) All gases consist of molecules. The molecules are rigid, elastic spheres identical in all respects for a given gas and different for different gases.

(ii) The size of a molecule is negligible compared with the average distance between two molecules.

(iii) The molecules are in a state of continuous random motion, moving in all directions with all possible velocities.

(iv) During the random motion, the molecules collide with one another and with the walls of the vessel.

- (v) The collisions are perfectly elastic and there are no forces of attraction or repulsion between them.
- (vi) Between two collisions, a molecule moves in a straight path with a uniform velocity.
- (vii) The collisions are almost instantaneous.
- (viii) The molecular density remains uniform throughout the gas.

9. **Pressure exerted by a gas.** According to kinetic theory of gases, the pressure exerted by a gas of mass M and volume V or density ρ is given by

$$P = \frac{1}{3} \frac{M}{V} \overline{v^2} = \frac{1}{3} \rho \overline{v^2} = \frac{1}{3} mn \overline{v^2}$$

Here n is the number of molecules per unit volume, m the mass of each molecule and $\overline{v^2}$ is the mean of square speed.

10. **Average kinetic energy of a gas.** Let M be the molecular mass and V the molar volume of a gas. Let m be the mass of each molecule. Then

- (i) Mean K.E. per mole of a gas,

$$E = \frac{1}{2} M \overline{v^2} = \frac{3}{2} PV = \frac{3}{2} RT = \frac{3}{2} k_B N_A T$$

- (ii) Mean K.E. per molecule of a gas,

$$\bar{E} = \frac{1}{2} m \overline{v^2} = \frac{3}{2} k_B T$$

This is the *kinetic interpretation of temperature*. That is, the temperature of a gas is a measure of the average kinetic energy of a molecule.

- (iii) K.E. of 1 gram of gas = $\frac{1}{2} \overline{v^2} = \frac{3}{2} \frac{RT}{M}$.

11. **Avogadro's law.** It states that equal volume of all gases under similar conditions of temperature and pressure contain equal number of molecules.

12. **Avogadro's number.** It is the number of particles present in one mole of a substance. Its most accepted value is

$$N_A = 6.0225 \times 10^{23} \text{ mole}^{-1}.$$

13. **Graham's law of diffusion.** It states that the rate of diffusion of a gas is inversely proportional to the square root of its density.

$$\frac{r_1}{r_2} = \sqrt{\frac{\rho_2}{\rho_1}}$$

14. **Dalton's law of partial pressures.** It states that the total pressure exerted by a mixture of non-reacting gases occupying a given volume is equal to the sum of the partial pressures which gas would exert if it alone occupied the same volume at the given temperature.

$$P = P_1 + P_2 + P_3 + \dots$$

15. **Average speed.** It is defined as the arithmetic mean of the speeds of the molecules of a gas at a given temperature.

$$\bar{v} = \frac{v_1 + v_2 + v_3 + \dots + v_n}{n}$$

$$\bar{v} = \sqrt{\frac{8k_B T}{\pi m}} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8PV}{\pi M}}$$

16. **Root mean square speed.** It is defined as the square root of the mean of the squares of the speeds of the individual molecules of a gas.

$$v_{rms} = \sqrt{\frac{v_1^2 + v_2^2 + v_3^2 + \dots + v_n^2}{n}}$$

$$v_{rms} = \sqrt{\frac{3k_B T}{m}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3PV}{M}}$$

17. **Most probable speed.** It is defined as the speed possessed by the maximum number of molecules in a gas sample at a given temperature.

$$v_{mp} = \sqrt{\frac{2k_B T}{m}} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2PV}{M}}$$

18. **Relations between $\bar{v}$, v_{rms} and v_{mp} .**

$$\bar{v} = 0.92 v_{rms}$$

$$v_{mp} = 0.816 v_{rms}$$

$$v_{rms} : \bar{v} : v_{mp} = 1.73 : 1.60 : 1.41$$

Clearly, $v_{rms} > \bar{v} > v_{mp}$.

19. **Degrees of freedom.** The degrees of freedom of a dynamical system are defined as the total number of coordinates or independent quantities required to describe completely the position and configuration of the system.

If N = number of particles in the system,
 k = number of independent relations between the particles

then the number of degrees of freedom of the system is

$$f = 2N - k.$$

A monoatomic molecule gas has 3 degrees of freedom, a diatomic gas molecule has 5 degrees of freedom. At high temperature, a diatomic molecule has 7 degrees of freedom.

20. **Law of equipartition of energy.** It states that in any dynamical system in thermal equilibrium, the energy of the system is equally divided amongst its various degrees of freedom and the energy associated with each degree of freedom is $\frac{1}{2} k_B T$, where k_B is Boltzmann's constant and T is the absolute temperature of the system.

21. Internal energies and specific heats of monoatomic, diatomic and polyatomic gases. The law of equipartition of energy leads to the following results :

- (i) For a gas of polyatomic molecules having f degrees of freedom,

Energy associated with 1 mole of gas,

$$U = \frac{f}{2} RT \quad C_v = \frac{f}{2} R,$$

$$C_p = \left(1 + \frac{f}{2}\right) R \quad \gamma = \frac{C_p}{C_v} = 1 + \frac{2}{f}$$

- (ii) For a monoatomic gas, $f = 3$, so

$$U = \frac{3}{2} RT, \quad C_v = \frac{3}{2} R,$$

$$C_p = \frac{5}{2} R, \quad \gamma = 1.66.$$

- (iii) For a diatomic gas with no vibrational mode, $f = 5$, so

$$U = \frac{5}{2} RT, \quad C_v = \frac{5}{2} R,$$

$$C_p = \frac{7}{2} R, \quad \gamma = 1.4.$$

- (iv) For a diatomic gas with vibrational mode, $f = 7$, so

$$U = \frac{7}{2} RT, \quad C_v = \frac{7}{2} R,$$

$$C_p = \frac{9}{2} R, \quad \gamma = \frac{9}{7} = 1.28.$$

- (v) For a triatomic gas of non-linear molecules, $f = 6$, so

$$U = 3RT, \quad C_v = 3R,$$

$$C_p = 4R, \quad \gamma = 1.33$$

- (vi) For a triatomic gas of linear molecules, $f = 7$, so

$$U = \frac{7}{2} RT, \quad C_v = \frac{7}{2} R,$$

$$C_p = \frac{9}{2} R, \quad \gamma = 1.28.$$

22. **Mean free path.** It is the average distance covered by a molecule between two successive collisions. It is given by

$$\bar{\lambda} = \frac{1}{\sqrt{2} n \pi d^2}$$

where n is the number density and d is the diameter of the molecule.

23. **Brownian motion.** It provides a direct evidence for the existence of molecules and their motion. The zig-zag motion of gas molecules is Brownian motion because it occurs due to random collision of molecules. But this motion cannot be seen. However, the zig-zag motion of pollen grains ($\approx 10^{-5}$ m) can be seen under a microscope.

IIT Entrance Exam

MULTIPLE CHOICE QUESTIONS WITH ONE CORRECT ANSWER

1. A vessel contains 1 mole of O_2 gas (relative molar mass 32) at a temperature T . The pressure of the gas is P . An identical vessel containing one mole of He gas (relative molar mass 4) at a temperature $2T$ has a pressure of

- (a) $P/8$ (b) P
(c) $2P$ (d) $8P$

[IIT 97]

2. The average translational kinetic energy of O_2 molecules (relative molar mass 32) at a particular temperature is 0.048 eV. The translational kinetic energy of N_2 molecules (relative molar mass 28) in eV at the same temperature is

- (a) 0.0015 (b) 0.003
(c) 0.048 (d) 0.768

[IIT 97]

3. The average translational energy and the r.m.s. speed of molecules in a sample of oxygen gas at 300 K

are 6.21×10^{-21} J and 484 ms^{-1} respectively. The corresponding values at 600 K are nearly (assuming ideal gas behaviour)

- (a) 12.42×10^{-21} J, 968 ms^{-1}
(b) 8.78×10^{-21} J, 684 ms^{-1}
(c) 6.21×10^{-21} J, 968 ms^{-1}
(d) 12.42×10^{-21} J, 684 ms^{-1}

[IIT 97]

4. A gas mixture consists of 2 moles of oxygen and 4 moles of argon at temperature T . Neglecting all vibrational modes, the total internal energy of the system is

- (a) $4RT$ (b) $9RT$
(c) $11RT$ (d) $15RT$

[IIT 99]

5. The ratio of the speed of sound in nitrogen gas to that in helium gas, at 300 K is

- (a) $\sqrt{2/7}$ (b) $\sqrt{1/7}$
(c) $(\sqrt{3})/5$ (d) $(\sqrt{6})/5$

[IIT 99]

Competition Section

Oscillations

GLIMPSES

1. **Periodic motion.** A motion which repeats itself over and over again after a regular interval of time is called a periodic motion.
2. **Oscillatory motion.** A motion in which a body moves back and forth repeatedly about a fixed point (called mean position) is called oscillatory or vibratory motion.
3. **Periodic function.** Any function that repeats its value at regular intervals of its argument is called a periodic function. The following sine and cosine functions are periodic with period T .

$$f(t) = \sin \frac{2\pi t}{T} \quad \text{and} \quad g(t) = \cos \frac{2\pi t}{T}$$

The periodic functions which can be represented by a sine or cosine curve are called *harmonic functions*. All harmonic functions are necessarily periodic but all periodic functions are not harmonic.

The periodic functions which cannot be represented by single sine or cosine function are called *non-harmonic functions*.

4. **Fourier theorem.** Two infinite sets of periodic functions with period T are

$$f_n(t) = \sin \frac{2\pi n t}{T}, \quad n = 1, 2, 3, 4, \dots$$

$$g_n(t) = \cos \frac{2\pi n t}{T}, \quad n = 0, 1, 2, 3, \dots$$

Fourier theorem states that any periodic function $F(t)$ with period T can be expressed as the unique combination of sine and cosine functions $f_n(t)$ and $g_n(t)$ with suitable coefficients. Mathematically,

$$F(t) = b_0 + \sum b_n \cos n\omega t + \sum a_n \sin n\omega t$$

where $\omega = 2\pi/T$. The coefficients $b_0, b_1, b_2, \dots; a_1, a_2, a_3, \dots$ are called *Fourier coefficients*. The special case of Fourier theorem in which only a_1 and b_1 are non-zero represents *simple harmonic motion* (S.H.M.).

$$F(t) = a_1 \sin \frac{2\pi t}{T} + b_1 \cos \frac{2\pi t}{T}$$

5. **Simple harmonic motion.** A particle is said to execute simple harmonic motion if it moves to and fro about a mean position under the action of a restoring force which is directly proportional to its displacement from the mean position and is always directed towards the mean position. If the displacement of the oscillating particle from the mean position is small, then

Restoring force $\propto$ Displacement

$$\text{or} \quad F \propto x$$

$$\text{or} \quad F = -kx$$

where k is a positive constant called *force constant* or *spring factor* and is defined as the restoring force produced per unit displacement. The negative sign shows that the restoring force always acts in the opposite direction of displacement x . The above equation defines SHM.

6. **Oscillation or cycle.** One complete back and forth motion of a particle is called cycle or vibration or oscillation.
7. **Displacement.** It is the distance of the oscillating particle from the mean position at any instant. It is denoted by x .
8. **Amplitude (A).** The maximum displacement of the oscillating particle on either side of its mean position is called its amplitude. Thus $x_{\max} = \pm A$.
9. **Time period.** It is the time taken by a particle to complete one oscillation about its mean position. It is denoted by T .
10. **Frequency.** It is the number of oscillations completed per second by a particle about its mean position. It is denoted by ν and is equal to the reciprocal of time period. Thus $\nu = \frac{1}{T}$

Frequency is measured in hertz.

$$1 \text{ hertz} = 1 \text{ Hz} = 1 \text{ oscillation per second} = 1 \text{ s}^{-1}$$

11. **Angular frequency.** It is the quantity obtained by multiplying frequency ν by a factor of 2π . It is denoted by ω .

$$\text{Thus } \omega = 2\pi\nu = \frac{2\pi}{T}$$

SI unit of $\omega = \text{rad s}^{-1}$.

12. **Phase.** The phase of vibrating particle at any instant gives the state of the particle as regards its position and the direction of motion at that instant. It is denoted by ϕ .
13. **Initial phase or epoch.** The phase of a vibrating particle corresponding to time $t = 0$ is called initial phase or epoch. It is denoted by ϕ_0 .
14. **Phase difference.** The phase difference between two vibrating particles tells the lack of harmony in the vibrating states of the two particles at any instant.
15. **Relation between SHM and uniform circular motion.** Simple harmonic motion is the projection of uniform circular motion upon a diameter of a circle. This circle is called the *reference circle* and the particle which revolves along it is called *reference particle* or *generating particle*.
16. **Displacement in SHM.** In a simple harmonic motion, the displacement of a particle from its equilibrium position at any instant t is given by

$$x(t) = A \cos(\omega t + \phi_0)$$

Here A is *amplitude* of the displacement, the quantity $(\omega t + \phi_0)$ is the *phase* of the motion and ϕ_0 is the *initial phase*.

When the time is measured from the mean position,

$$x(t) = A \sin \omega t$$

When the time is measured from the extreme position,

$$x(t) = A \cos \omega t$$

The angular frequency ω , frequency ν and time period T of the motion are given by

$$\omega = \sqrt{\frac{a}{x}} \quad \text{or} \quad \omega = \sqrt{\frac{k}{m}}$$

$$\nu = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{a}{x}} \quad \text{or} \quad \nu = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$T = 2\pi \sqrt{\frac{\text{Displacement}}{\text{Acceleration}}} = 2\pi \sqrt{\frac{x}{a}}$$

$$\text{or} \quad T = 2\pi \sqrt{\frac{\text{Inertia factor}}{\text{Spring factor}}} = 2\pi \sqrt{\frac{m}{k}}$$

17. **Velocity in SHM.** It is the rate of change of displacement of the particle at any instant. It is given by

$$\begin{aligned} v &= \frac{dx}{dt} = \frac{d}{dt} [A \cos(\omega t + \phi_0)] \\ &= -\omega A \sin(\omega t + \phi_0) = -\omega \sqrt{A^2 - x^2} \end{aligned}$$

The maximum value of velocity is called *velocity amplitude* v_m of the motion.

$$\text{Thus } v_m = \omega A = \frac{2\pi}{T} A$$

At the mean position, particle velocity $= v_m = \omega A$

At the extreme position, particle velocity $= 0$.

18. **Acceleration in SHM.** It is the rate of change of velocity of the particle at any instant. It is given by

$$\begin{aligned} a &= \frac{dv}{dt} = \frac{d}{dt} [-\omega A \sin(\omega t + \phi_0)] \\ &= -\omega^2 A \cos(\omega t + \phi_0) = -\omega^2 x \end{aligned}$$

$$\text{i.e., } a \propto x$$

The maximum value of acceleration of particle is called *acceleration amplitude* a_m . Thus

$$a_m = \omega^2 A$$

At the mean position, particle acceleration $= 0$

At the extreme position, particle acceleration,

$$a_m = \omega^2 A$$

19. **Phase relationship between displacement, velocity and acceleration.** In SHM, the particle velocity is ahead of displacement by $\pi/2$ rad while acceleration is ahead of displacement by π rad.
20. **Energy of SHM.** If a particle of mass m executes SHM, then at a displacement x from mean position, the particle possesses potential and kinetic energy. At any displacement x ,

$$\text{Potential energy, } U = \frac{1}{2} m \omega^2 x^2 = \frac{1}{2} k x^2$$

$$\begin{aligned} \text{Kinetic energy, } K &= \frac{1}{2} m \omega^2 (A^2 - x^2) \\ &= \frac{1}{2} k (A^2 - x^2) \end{aligned}$$

Total energy,

$$E = U + K = \frac{1}{2} m \omega^2 A^2 = 2\pi^2 m \nu^2 A^2$$

If there is no friction, the total mechanical energy, $E = K + U$, of the system always remains constant even though K and U change.

21. **Motion of a massless loaded spring.** When a mass m is attached to a massless spring and pulled downwards, it executes SHM. If l is extension in the spring on attaching mass m and k is its force constant, then time period of SHM executed by the spring

$$T = 2\pi \sqrt{\frac{l}{g}} = 2\pi \sqrt{\frac{m}{k}}$$

22. **Spring cut into parts.** If we divide the spring of spring constant k into n equal parts, the spring constant of each part becomes nk . Hence the time period when the same mass m is suspended from each part is

$$T = 2\pi \sqrt{\frac{m}{nk}}$$

23. **Springs connected in series.** If two springs of spring constants k_1 and k_2 are connected in series, then the spring constant k of the combination is given by

$$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} \quad \text{or} \quad k = \frac{k_1 k_2}{k_1 + k_2}$$

$$\therefore T = 2\pi \sqrt{\frac{m(k_1 + k_2)}{k_1 k_2}}$$

24. **Springs connected in parallel.** If two springs of spring constants k_1 and k_2 are connected in parallel, then the spring constant k of combination is

$$k = k_1 + k_2 \quad \therefore T = 2\pi \sqrt{\frac{m}{k_1 + k_2}}$$

25. **Simple pendulum.** A simple pendulum is a heavy point mass suspended by a weightless, inextensible and a perfectly flexible string from a rigid support about which it can vibrate freely. The distance between the point of suspension and the point of oscillation is called *length of the pendulum* (l). When the metallic bob is displaced from mean position, it executes SHM.

$$\text{Time period, } T = 2\pi \sqrt{\frac{l}{g}}$$

26. **Second's pendulum.** A second's pendulum is a pendulum whose time period is two seconds. Its length is 99.3 cm.
27. **Motion of a liquid in a U-tube.** When a liquid of density ρ and contained in a U-tube upto height h is depressed, it executes SHM of time period,

$$T = 2\pi \sqrt{\frac{h}{g}}$$

28. **Motion of a body dropped in a tunnel dug along the diameter of earth.** When a body is dropped in a tunnel dug along the diameter of the earth, it executes SHM. If R is radius of the earth, then its time period is

$$T = 2\pi \sqrt{\frac{R}{g}}$$

29. **Motion of a body floating in a liquid.** When a body made of material of density ρ and total vertical length L floats in a liquid of density ρ , such

that its length h is submerged in the liquid, it executes SHM on being pushed into the liquid.

$$T = 2\pi \sqrt{\frac{\rho L}{\sigma g}} = 2\pi \sqrt{\frac{h}{g}}$$

30. **Free oscillations.** If a body, capable of oscillation, is slightly displaced from its position of equilibrium and then released, it starts oscillating with a frequency of its own. Such oscillations are called free oscillations. The frequency with which a body oscillates is called *natural frequency* and is given by

$$v_0 = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

Here a body continues to oscillate with constant amplitude and fixed frequency.

31. **Damped oscillations.** The oscillations in which amplitude decreases gradually with the passage of time are called damped oscillations.

The energy of a real oscillator decreases because a part of its mechanical energy is used in doing work against the frictional forces and is lost as heat. If the *damping force* is given by $F_d = -bv$, where v is the velocity of the oscillator and b is a *damping constant*, then the displacement of the oscillator is given by,

$$x(t) = Ae^{-bt/2m} \cos(\omega't + \phi)$$

where ω' , the angular frequency of the damped oscillator, is given by

$$\omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$

If the damping constant is small then $\omega' \approx \omega$, where ω is the angular frequency of the undamped oscillator. The mechanical energy E of the oscillator is given by

$$E(t) = \frac{1}{2} kA^2 e^{-bt/m}$$

32. **Forced oscillations.** When a body oscillates under the influence of an external periodic force, not with its own natural frequency but the frequency of the external periodic force, its oscillations are said to be forced oscillations.
33. **Resonant oscillations.** It is a particular case of forced oscillations in which the frequency of the driving force is equal to the natural frequency of the oscillator itself and the amplitude of oscillations is greatest. Such oscillations are called resonant oscillations and phenomenon is called *resonance*.
34. **Coupled oscillations.** A system of two or more oscillators linked together in such a way that there is mutual exchange of energy between them is called a coupled oscillator. The oscillations of such a system are called coupled oscillations.

Competition Section

Waves

GLIMPSES

1. **Wave motion.** It is a kind of disturbance which travels through a medium due to the repeated vibrations of the particles of the medium about their mean positions, the disturbance being handed over from one particle to the next. In a wave both information and energy propagate from one point to another but there is no motion of matter as a whole through a medium.
2. **Three basic types of waves.**
 - (i) **Mechanical waves.** The waves which require a mechanical medium for their propagation are called mechanical waves or elastic waves. For their propagation, the medium must possess the properties of inertia and elasticity. For example, water waves, sound waves, etc.
 - (ii) **Electromagnetic waves.** The waves which travel in the form of oscillating electric and magnetic fields are called electromagnetic waves. Such waves do not require any material medium for their propagation. For example, visible light, radio waves, etc.
 - (iii) **Matter waves.** The waves associated with microscopic particles, such as electrons, protons, neutrons, atoms, molecules, etc., when they are in motion, are called matter waves or de-Broglie waves. Matter waves associated with fast moving electrons are used in electron microscopes.
3. **Spring-model for the propagation of a wave through an elastic medium.** Energy transfer takes place because of the coupling through elastic forces between neighbouring oscillating parts of the medium.
4. **Transverse waves.** These are the waves in which particles of the medium vibrate about their mean positions in a direction perpendicular to the direction of propagation of the disturbance. These waves can propagate in those media which have a shear modulus of elasticity e.g., solids.
5. **Longitudinal waves.** These are the waves in which particles of the medium vibrate about their mean positions along the direction of propagation of the disturbance. These waves can propagate in those media having a bulk modulus of elasticity and are therefore possible in all media : solids, liquids and gases.
6. **Progressive wave.** A wave that moves from one point of medium to another is called a progressive wave.
7. **Amplitude (A).** It is the maximum displacement suffered by the particles of the medium from the mean position during the propagation of a wave.
8. **Time period (T).** It is the time in which a particle of the medium completes one vibration about its mean position.
9. **Frequency (v).** It is the number of waves produced per second in a given medium.
10. **Wavelength (λ).** It is the distance covered by a wave during the time a particle of the medium completes one vibration about its mean position. It is the distance between two nearest particles of the medium which are vibrating in the same phase.
11. **Angular wave number or propagation constant.** It represents the phase change per unit distance (or per unit path difference). It is equal to $2\pi/\lambda$.
Thus $k = \frac{2\pi}{\lambda}$
The SI unit of k is radian per metre (rad m^{-1}).
12. **Wave velocity (v).** It is the distance travelled by a wave in one second.
13. **Relation between wave velocity, frequency and wavelength.**
Wave velocity = Frequency $\times$ wavelength
or $v = v \lambda$

14. Relation between wave velocity, time period and wavelength.

$$\text{Wave velocity} = \frac{\text{Wavelength}}{\text{Time period}}$$

or $v = \frac{\lambda}{T} = \frac{\omega}{k}, \quad k = \frac{2\pi}{\lambda}$

15. Velocity of transverse waves.

(i) Velocity of transverse waves in a solid of modulus of rigidity η and density ρ is given by $v = \sqrt{\frac{\eta}{\rho}}$

(ii) Velocity of transverse waves in a string of mass per unit length m and stretched under tension T is given by $v = \sqrt{\frac{T}{m}}$

16. Velocity of longitudinal waves.

(i) Velocity of longitudinal waves in an extended solid (earth's crust) of bulk modulus κ , modulus of rigidity η and density ρ is given by

$$v = \sqrt{\frac{\kappa + \frac{4}{3}\eta}{\rho}}$$

The factor $\kappa + (\frac{4}{3})\eta$ is called elongational elasticity.

(ii) Velocity of longitudinal waves in a long rod of Young's modulus Y and density ρ is given by

$$v = \sqrt{\frac{Y}{\rho}}$$

(iii) Velocity of longitudinal waves in a liquid of bulk modulus κ and density ρ is given by

$$v = \sqrt{\frac{\kappa}{\rho}}$$

(iv) Velocity of longitudinal waves in a gaseous medium of bulk modulus κ and density ρ is given by

$$v = \sqrt{\frac{\kappa}{\rho}}$$

According to Newton, when sound travels through gas, the changes taking place in the medium are isothermal in nature. So Newton's formula for the speed of sound is

$$v = \sqrt{\frac{\kappa_{\text{iso}}}{\rho}} = \sqrt{\frac{P}{\rho}}, \text{ where } P = \text{pressure of the gas.}$$

According to Laplace, when sound travels in a gas, the changes taking place in the medium are adiabatic. So Laplace formula for the speed of sound is

$$v = \sqrt{\frac{\kappa_{\text{adia}}}{\rho}} = \sqrt{\frac{\gamma P}{\rho}}$$

where $\gamma = \frac{C_p}{C_v}$ = specific heat ratio.

17. Factors affecting velocity of sound through gases

(i) Effect of pressure. Pressure has no effect on the speed of sound in a gas.

(ii) Effect of density. $v \propto \frac{1}{\sqrt{\rho}}$ or $\frac{v_1}{v_2} = \sqrt{\frac{\rho_2}{\rho_1}}$

(iii) Effect of temperature. As $v = \sqrt{\frac{\gamma R T}{M}}$

$$\therefore v \propto \sqrt{T} \quad \text{or} \quad \frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}}$$

Temperature coefficient of velocity of sound is given by

$$\alpha = \frac{v_t - v_0}{t}$$

For air, $\alpha = 0.61 \text{ ms}^{-1} \text{ } ^\circ\text{C}^{-1}$.

(iv) Effect of humidity. Sound travels faster in moist air.

(v) Effect of wind. If the wind blows with velocity w in a direction making an angle θ with the direction of sound, then the resultant velocity of sound will be $v' = v + w \cos \theta$

18. Wave equation. A plane progressive harmonic wave travelling along positive X-direction may be represented as

(i) $y = A \sin (\omega t - kx)$, where $k = 2\pi / \lambda$

(ii) $y = A \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$

(iii) $y = A \sin \frac{2\pi}{T} (vt - x)$

where v is the wave-velocity, λ the wavelength, A the amplitude of the oscillating particles of the medium and y is the displacement of the particle located at position x at any instant t .

If the wave is travelling along negative X-direction, the minus sign is replaced by plus sign in the above equations. Thus

(i) $y = A \sin (\omega t + kx)$, where $k = 2\pi / \lambda$

(ii) $y = A \sin 2\pi \left(\frac{t}{T} + \frac{x}{\lambda} \right)$

(iii) $y = A \sin \frac{2\pi}{\lambda} (vt + x)$

19. Phase and phase difference. Phase is the argument of the sine or cosine function representing the wave. Thus phase,

$$\phi = 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$$

The relation between phase difference ($\Delta\phi$) and time interval Δt is

$$\Delta\phi = \frac{2\pi}{T} \Delta t$$

Thus in time period T , the phase of a particle changes by 2π .

The relation between phase difference ($\Delta\phi$) and path difference (Δx) is

$$\Delta\phi = -\frac{2\pi}{\lambda} \Delta x$$

The negative sign indicates that farther the particle is located from the origin in the positive X -direction, the more it lags behind in phase. Clearly, the phase difference between two particles located at separation λ is 2π .

20. **Principle of superposition of waves.** When a number of waves travel through a medium simultaneously, the resultant displacement at any point of the medium is equal to the vector sum of the displacements of the individual waves. If $\vec{y}_1, \vec{y}_2, \vec{y}_3, \dots, \vec{y}_n$ are the displacements of n waves superposing each other at a point, then the resultant displacement at that point will be

$$\vec{y} = \vec{y}_1 + \vec{y}_2 + \vec{y}_3 + \dots + \vec{y}_n$$

21. **Reflection of a wave.** When a wave is reflected from a rigid boundary or a closed end, it is reflected back with a phase reversal or phase difference of π radians but reflection at an open boundary takes place without any phase change.

22. **Stationary waves.** When two progressive waves of equal amplitude and frequency, travelling in opposite directions along a straight line superpose each other, the resultant wave does not travel in either direction and is called a *stationary* or *standing wave*. At some points, the particles of the medium always remain at rest. These are called *nodes*. At some other points, the amplitude of oscillation is maximum. These are called *antinodes*.

Consider a plane progressive harmonic wave travelling along positive X -direction.

$$y_1 = A \sin(\omega t - kx) \quad (\text{incident wave})$$

If this wave is reflected from a *free end*, then

$$y_2 = A \sin(\omega t + kx) \quad (\text{reflected wave})$$

The stationary wave formed by the superposition of the incident and reflected waves will be

$$y = y_1 + y_2 = 2A \cos kx \sin \omega t$$

In this case, the points $x = 0, \lambda/2, 3\lambda/2, \dots$ will be antinodes and the points $x = \lambda/4, 3\lambda/4, 5\lambda/4, \dots$ will be nodes.

If the wave is reflected from a *rigid end*, then

$$y_2 = -A \sin(\omega t + kx)$$

The equation of the stationary wave will be

$$y = -2A \sin kx \cos \omega t$$

In this case, the points $x = 0, \lambda/2, \lambda, 3\lambda/2, \dots$ will be nodes and the points $x = \lambda/4, 3\lambda/4, 5\lambda/4, \dots$ will be antinodes.

Separation between two successive nodes or antinodes $= \lambda/2$

Separation between a node and nearest antinode $= \lambda/4$

23. **Modes of vibrations of strings.** On a stretched string, transverse stationary waves are formed due to superposition of direct and the reflected transverse waves.

For fundamental mode :

$$\lambda_1 = 2L, \quad v_1 = \frac{v}{\lambda_1} = \frac{1}{2L} \sqrt{\frac{T}{m}} = v \text{ (say)}$$

For second mode :

$$\lambda_2 = L, \quad v_2 = 2v \quad (\text{second harmonic or first overtone})$$

For third mode :

$$\lambda_3 = 2L/3, \quad v_3 = 3v \quad (\text{third harmonic or second overtone})$$

For p th mode : When the string vibrates in p loops,

$$v_p = \frac{p}{2L} \sqrt{\frac{T}{m}} = pv \quad [p\text{th harmonic or } (p-1)\text{th overtone}]$$

Fundamental frequency for a string of diameter D and density ρ ,

$$v = \frac{1}{LD} \sqrt{\frac{T}{\pi \rho}}$$

24. **Laws of transverse vibrations of a stretched string.** The fundamental frequency of a vibrating is

(i) inversely proportional to its length.

$$v \propto \frac{1}{L} \quad (\text{Law of length})$$

(ii) directly proportional to the square root of its tension.

$$v \propto \sqrt{T} \quad (\text{Law of tension})$$

(iii) inversely proportional to its mass per unit length.

$$v \propto \frac{1}{\sqrt{m}} \quad (\text{Law of mass})$$

Combining all the factors, we get

$$v = \frac{1}{2L} \sqrt{\frac{T}{m}} = \frac{1}{LD} \sqrt{\frac{T}{\pi \rho}}$$

where ρ is the density and D the diameter of the string.

25. **Organ pipe.** It is the simplest musical instrument in which sound is produced by setting an air column into vibrations. Longitudinal stationary waves are formed on account of superposition of incident and reflected longitudinal waves.

26. **Modes of vibrations of closed organ pipes.** Longitudinal stationary waves are formed in an organ pipe closed at one end.

For fundamental mode

$$\lambda_1 = 4L, \quad v_1 = \frac{v}{\lambda_1} = \frac{1}{4L} \sqrt{\frac{\gamma P}{\rho}} = v \text{ (say)}$$

For second mode : $\lambda_2 = 4L/3, \quad v_2 = 3v$
(third harmonic or first overtone)

For third mode : $\lambda_3 = 4L/5, \quad v_3 = 5v$
(fifth harmonic or second overtone)

For p th mode :

$$\lambda_p = 4L/(2p-1), \quad \lambda_p = (2p-1)v$$

[($2p-1$)th harmonic or ($p-1$)th overtone]

Here $v_1 : v_2 : v_3 : v_4 : \dots = 1 : 3 : 5 : 7 : \dots$
(only odd harmonics)

27. **Modes of vibrations of open organ pipe.** Anti-nodes are formed at both ends, separated by one node in the fundamental mode.

For fundamental mode:

$$\lambda'_1 = 2L, \quad v'_1 = \frac{v}{2L} = \frac{1}{2L} \sqrt{\frac{\gamma P}{\rho}} = v \text{ (say)}$$

For second mode : $\lambda'_2 = L, \quad v'_2 = 2v$
(second harmonic or first overtone)

For third mode : $\lambda'_3 = 2L/3, \quad v'_3 = 3v$
(third harmonic or second overtone)

For p th mode $\lambda'_p = 2L/p, \quad v'_p = pv$
[p th harmonic or ($p-1$)th overtone]

Here $v_1 : v_2 : v_3 : v_4 : \dots = 1 : 2 : 3 : 4 : \dots$
(Both odd and even harmonics)

28. **Resonance tube.** It is an organ pipe closed at one end. If L_1 and L_2 be the first and second resonance lengths with a tuning fork of frequency v , then the speed of sound in air is given by

$$v = 4v(L_1 + 0.3D),$$

D = internal diameter of resonance tube

or $v = 2v(L_2 - L_1)$

End correction = $0.3D = \frac{L_2 - 3L_1}{2}$

29. **Vibrations in rods clamped in the middle.** Here antinodes are formed at the ends and a node in the middle.

Fundamental frequency, $v_1 = \frac{v}{2L} = v$

First overtone or third harmonic, $v_2 = \frac{3v}{2L} = 3v$

Second overtone or fifth harmonic, $v_3 = \frac{5v}{2L} = 5v$

30. **Beats.** The periodic variations in the intensity of sound due to the superposition of two sound waves of slightly different frequencies are called beats. One rise and one fall of intensity constitute one beat. The number of beats produced per second is called beat frequency.

Beat frequency, $v_{\text{beat}} = v_1 - v_2$

Beats may be used to determine the frequency of a tuning fork. It must be noted here that

- When the prong of a tuning fork is slightly loaded with wax, its frequency of vibration decreases.
- When the prong of a tuning fork is filed slightly, its frequency of vibration increases.

31. **Doppler effect in sound.** The phenomenon of the change in apparent pitch of sound due to relative motion between the source of sound and the observer is called Doppler effect. If v, v_0, v_s and v_m are the velocities of sound, observer, source and medium (in the direction of sound) respectively, then the apparent frequency is given by

$$v' = \frac{v + v_m - v_0}{v + v_m - v_s} \times v$$

For the medium at rest ($v_m = 0$),

$$v' = \frac{v - v_0}{v - v_s} \times v$$

Here the velocities are taken *positive* in the *source to observer* ($S \rightarrow O$) direction and *negative* in the opposite ($O \rightarrow S$) direction.

Special cases :

- When the source moves towards the stationary observer,

$$v' = \frac{v}{v - v_s} \times v \quad (v' > v)$$

- When the source moves away from the stationary observer,

$$v' = \frac{v}{v + v_s} \times v \quad (v' < v)$$

- When the observer moves towards the stationary source,

$$v' = \frac{v + v_0}{v} \times v \quad (v' > v)$$

- When the observer moves away from the stationary source,

$$v' = \frac{v - v_0}{v} \times v \quad (v' < v)$$

- When both source and observer move towards each other,

$$v' = \frac{v + v_0}{v - v_s} \times v \quad (v' > v)$$

- (vi) When both source and observer move away from each other,

$$v' = \frac{v - v_0}{v + v_s} \times v \quad (v' < v)$$

- (vii) When source moves towards observer and observer away from the source,

$$v' = \frac{v - v_0}{v - v_s} \times v$$

- (viii) When source moves away from observer and observer towards the source,

$$v' = \frac{v + v_0}{v + v_s} \times v$$

32. **Musical sound.** A musical sound consists of quick, regular and periodic succession of compressions and rarefactions without a sudden change in amplitude. It produces pleasing effect on the ears.

33. **Noise.** A noise consists of slow, irregular and non periodic succession of compressions and rarefactions, that may have a sudden change in amplitude. It produces non-pleasing effect on the ears.

34. **Intensity of sound (I).** The intensity of sound at any point may be defined as the amount of sound energy passing per unit time per unit area around that point in a perpendicular direction.

35. **Zero level or threshold of hearing (I_0).** The lowest intensity of sound that can be perceived by the human ear is called threshold of hearing. For a sound of frequency 1 kHz, it is found that the threshold of hearing is 10^{-12} Wm^{-2} .

36. **Characteristics of a musical sound.** These are

- Pitch.** It is the characteristic of musical sound that helps the listener in distinguishing a shrill note from a grave (flat or dull) one. It depends on frequency.
- Quality.** It is the characteristic of the musical sound that distinguishes between two sounds of same pitch and loudness from one another. It depends on the number or intensity of overtones.
- Loudness.** The sensation of hearing which enables us to distinguish between a loud and a faint sound, is called loudness. It depends on intensity.

37. **Units of loudness.** The unit of loudness of a sound is **bel**. The loudness of a sound is said to be 1 bel, if its intensity is 10 times that of the threshold of hearing.

The loudness of a sound of intensity I is given by

$$L = \log_{10} \frac{I}{I_0},$$

where I_0 is threshold of hearing. It is called **Weber-Fechner law**.

The practical and a smaller unit of loudness is **decibel (dB)**.

$$1 \text{ decibel} = \frac{1}{10} \text{ bel}$$

In decibels, the loudness of a sound of intensity I is given by

$$L = 10 \log_{10} \frac{I}{I_0}$$

38. **Reverberation of sound.** It is the phenomenon of persistence of sound after the source has stopped producing sound. The time for which sound persists after the source has stopped producing sound is called **reverberation time (T)**.

According to **Sabine law**, reverberation time of a hall is given by

$$T = \frac{0.16 V}{\sum a_i s_i}$$

where V is volume of the hall and $\sum a_i s_i = a_1 s_1 + a_2 s_2 + \dots$ is the total absorption of the hall. Here $s_1, s_2, \dots$ are the surface areas of materials with absorption coefficients $a_1, a_2, \dots$ respectively.

39. **Interval between two notes.** The ratio of the frequencies of two notes is called interval between them. Two notes are said to be in **unison** if their frequencies are equal, that is interval between them is 1 : 1. Some other common intervals found useful in producing musical sound are as follows :

- | | |
|------------------------|------------------------|
| (a) octave (1 : 2) | (b) majortone (8 : 9) |
| (c) minortone (9 : 10) | (d) semitone (15 : 16) |

The interval between any two notes is obtained by multiplying the various intervening intervals.

40. **Musical scale.** A series of notes arranged such that their fundamental frequencies have definite ratios is called a musical scale.

41. **Major diatonic scale.** The eight notes of major diatonic scale starting with 256 as keynote along with their Helmholtz and Indian notation are as below :

Helmholtz notation :	C	D	E	F	G	A	B	C
Indian notation :	Sa	Re	Ga	Ma	Pa	Dha	Ni	Sa
Frequency of notes with 256 as keynote :	256	288	320	$341 \frac{1}{3}$	384	$426 \frac{2}{3}$	480	512
Intervals :	9/8	10/9	16/15	9/8	10/9	9/8	16/15	